

Thermohydraulic and Thermodynamic Performance of a Parabolic Trough Solar Collector Receiver Equipped with a Central Hollow Twisted Square Promoter and Hybrid Nanofluids

1. Introduction

1.1. Background

The continuous growth of industrial activity, urban development, and commercial demand has placed increasing pressure on global energy systems. For many decades, this demand has been met largely through fossil-fuel-based resources such as coal, oil, and natural gas. Although these resources have supported economic and industrial development, their large-scale consumption has also intensified environmental problems through the release of carbon dioxide and other atmospheric pollutants [1]. These emissions are associated with climate change, air-quality deterioration, and adverse impacts on human health. Therefore, reducing dependence on fossil fuels has become both an environmental necessity and a technological challenge.

One of the clearest indicators of this challenge is the continued increase in atmospheric greenhouse gases. Olia et al. [2] reported that the average atmospheric concentration of CO₂ increased from about 390 ppm in 2011 to approximately 410.19 ppm in 2018. This rise reflects the scale of the global emissions problem and supports the urgent need for cleaner energy systems. Renewable energy resources, including hydropower, biomass, geothermal energy, solar energy, wind energy, and wave and tidal energy, offer important pathways for reducing the environmental impact of energy production [2,4]. Among these alternatives, solar energy is especially attractive because it is abundant, widely distributed, and capable of producing both heat and electricity.

Solar energy represents one of the largest renewable energy resources available to humanity. Lewis and Nocera [5] emphasized that the solar energy reaching the Earth within one hour can exceed the total energy consumed by humanity in an entire year. In a recent review, Kim et al. [3] also highlighted the large scale of the solar resource, noting that only 0.016% of the Earth's daily solar energy flux could theoretically meet global energy demand based on 2022 consumption levels. These figures show that the main challenge is not the availability of solar energy itself, but the efficiency, reliability, and cost-effectiveness of the technologies used to collect and convert it.

Solar energy technologies may be classified according to the type of system and the form of useful energy produced. In general, they can be passive or active systems. Passive systems rely mainly on natural collection and heat-transfer mechanisms, while active systems require external

components such as pumps, tracking mechanisms, or electronic control units. Solar technologies may also be grouped according to the useful output they provide. Photovoltaic systems convert sunlight directly into electricity, whereas solar thermal systems absorb solar radiation and convert it into useful heat [1,2] as shown in figure 1.

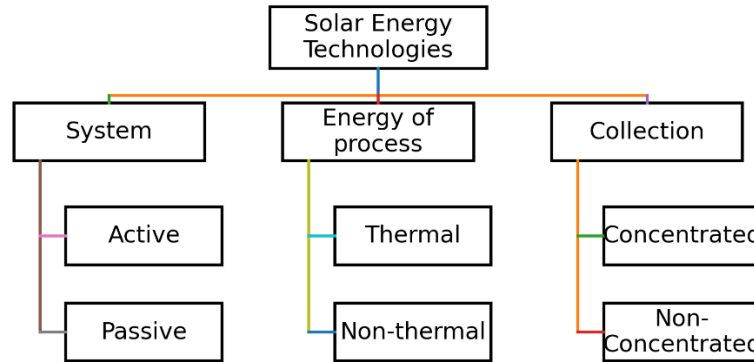


Figure 1: Classification of solar energy technologies according to system type and useful energy output [1].

Each solar-conversion route has its own advantages and limitations. Photovoltaic systems have developed rapidly, and their costs have decreased over time. However, the conversion efficiency of conventional photovoltaic cells is often reported within the range of 10-20%, and their performance may decline when cell temperature increases [2,6]. Solar thermal technologies, in contrast, are particularly suitable for applications where useful heat is required, including water heating, cooling, desalination, industrial process heating, and power generation through thermodynamic cycles.

In solar thermal systems, absorbed radiation is transferred as heat to a working fluid such as water, thermal oil, molten salt, air, nitrogen, or helium [2,7]. The performance of these systems depends strongly on how effectively the absorbed solar energy is transferred to the working fluid and delivered to the intended process. Concentrating solar thermal collectors are particularly important because they can achieve higher operating temperatures than non-concentrating collectors by focusing direct solar radiation onto a smaller receiver area [1,3]. Among these technologies, parabolic trough solar collectors have received significant attention because of their technical maturity and suitability for medium- and high-temperature applications. Improving their thermal performance remains essential for increasing useful energy output, reducing losses, and supporting wider deployment of solar thermal energy systems.

1.2. Solar Collectors

Solar collectors are thermal devices used to capture solar radiation and convert it into useful heat. The collected energy is usually transferred to a working fluid, which then delivers the heat to a process, a storage unit, or a power-generation cycle. The choice of collector depends mainly on the required operating temperature, the concentration level, the tracking method, and the final application. For this reason, solar collectors are commonly classified into stationary collectors and sun-tracking concentrating collectors [8].

Stationary collectors, also known as non-tracking collectors, absorb solar radiation without continuously following the apparent motion of the Sun. This category includes flat-plate collectors, evacuated-tube collectors, compound parabolic collectors, bowl collectors, through-pass air collectors, and unglazed transpired collectors [8]. These collectors are generally simple, mechanically reliable, and suitable for low- and medium-temperature applications such as water heating, space heating, crop drying, and air heating. Their main limitation is that they do not strongly concentrate solar radiation, which restricts their ability to reach high operating temperatures. Sun-tracking concentrating collectors are designed to increase the solar flux incident on the receiver. This is achieved by reflecting or refracting radiation from a relatively large aperture area onto a smaller receiver area using mirrors or lenses. Depending on their optical configuration, these collectors may concentrate radiation along a focal line or onto a focal point. One-axis tracking collectors include parabolic trough collectors, cylindrical trough collectors, and linear Fresnel collectors, while two-axis tracking collectors include parabolic dish collectors, solar tower systems, and circular Fresnel systems [8,12]. Figure 2 summarizes the general classification of solar collectors.

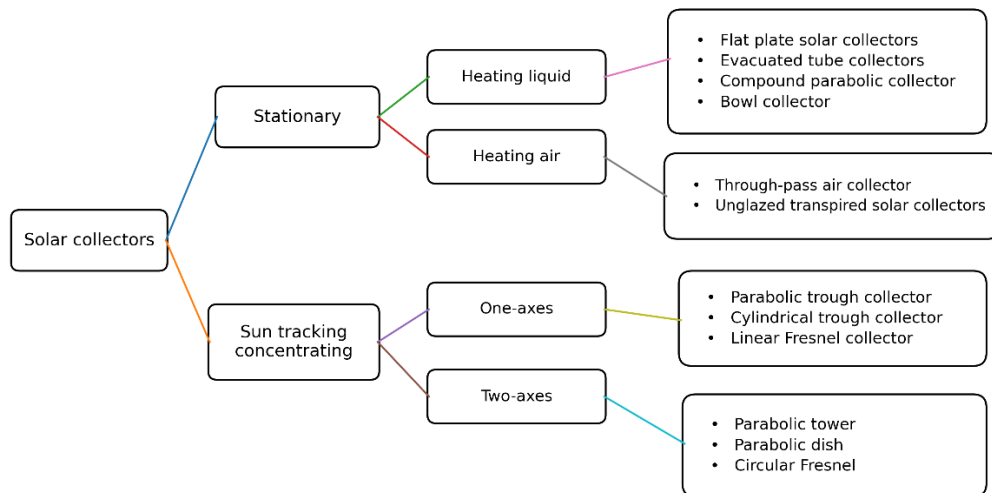


Figure 2: Classification of solar collectors into stationary and sun-tracking concentrating systems [2].

A key parameter in concentrating solar collectors is the concentration ratio. It is commonly defined as the ratio between the concentrator aperture area and the receiver area, as expressed in Eq. (1.1) [8,17]:

$$CR = \frac{A_c}{A_r} \quad (1.1)$$

where CR is the concentration ratio, A_c is the concentrator aperture area, and A_r is the receiver area. A concentration ratio greater than one indicates that solar radiation is being concentrated. Higher concentration ratios allow the collector to reach higher temperatures, but they also increase the need for accurate optical alignment and tracking.

In industrial solar thermal applications, concentrating solar collectors are mainly represented by four configurations: parabolic trough solar collectors, linear Fresnel reflectors, solar towers, and parabolic dish collectors [9-12]. These technologies differ in their focus type, receiver arrangement, tracking requirement, concentration range, and practical applications. A concise comparison is presented in Table 1.

Table 1: Concise comparison of major concentrating solar collector technologies. Adapted from Kalogirou [8], IRENA [9], IEA [10], and NREL [12].

Technology	Focus type	Tracking	Typical concentration and temperature level	Main strengths and limitations
Parabolic trough solar collector	Line focus	Single-axis	Medium concentration, commonly below 100; suitable for medium to high temperatures	Mature technology, widely deployed, and suitable for industrial heat and power generation. Receiver heat loss and working-fluid limitations remain important challenges.
Linear Fresnel reflector	Line focus	Single-axis	Medium concentration, commonly below 100	Structurally simpler than PTSCs and can support direct steam generation. Optical losses are usually higher because of shading, blocking, and cosine effects.
Solar tower	Point focus	Two-axis heliostat field	High concentration, can exceed 1000	Suitable for high-temperature and large-scale power generation. Requires many heliostats, accurate tracking, and more complex control.
Parabolic dish collector	Point focus	Two-axis	High concentration, can exceed 1000	Capable of high temperatures and high optical concentration. Often used with Stirling engines, but large-scale deployment can be limited by cost and modularity.

Among these technologies, parabolic trough solar collectors and linear Fresnel reflectors are line-focus systems, whereas solar towers and parabolic dishes are point-focus systems. This distinction is important because it affects the concentration ratio, achievable temperature, optical complexity, and receiver design. In general, line-focus systems are less complex and more suitable for medium-temperature applications, while point-focus systems can achieve higher temperatures but require more accurate tracking and control [8,10,11].

Parabolic trough solar collectors have become one of the most established concentrating solar technologies. They use a parabolic reflector to focus direct solar radiation onto a receiver tube located along the focal line. Their single-axis tracking arrangement is less complex than the two-axis tracking required in solar towers and parabolic dish systems. This makes PTSCs suitable for applications such as industrial process heat, solar refrigeration, air-conditioning, steam generation, and solar thermal power plants [13,14].

The strong interest in PTSCs is also related to their commercial maturity. IRENA [9] reported that parabolic trough technology has held the largest share of the global concentrating solar thermal market compared with other concentrating technologies. This maturity makes PTSCs a practical platform for further research and development. However, their performance still depends strongly on receiver heat transfer, heat loss reduction, and the thermophysical behavior of the working fluid. These aspects justify the focus of the present thesis on improving the thermal performance of PTSC receiver systems.

1.3. Parabolic Trough Solar Collector (PTSC)

A parabolic trough solar collector (PTSC) is a line-focus concentrating solar collector that uses a curved reflective surface to concentrate direct solar radiation onto a receiver tube located along the focal line of the parabola. The concentrated radiation is absorbed by the receiver and transferred as heat to a working fluid flowing inside the absorber tube. This thermal energy can then be used directly for industrial heating or indirectly for power generation through a thermodynamic cycle [1,8]. In the literature, the terms PTC and PTSC are often used interchangeably; however, this proposal uses PTSC for consistency.

At its core, a PTSC operates by bringing together two distinct physical processes: the optical concentration of sunlight and the convective removal of the resulting heat. The curved reflector focuses incoming direct beam radiation onto the receiver, where the selective absorber coating converts the concentrated flux into thermal energy at the tube wall. The heat is then transferred through the absorber wall to the heat transfer fluid (HTF), which transports the useful energy to a storage unit, heat exchanger, steam-generation system, or process load as in figure 3.

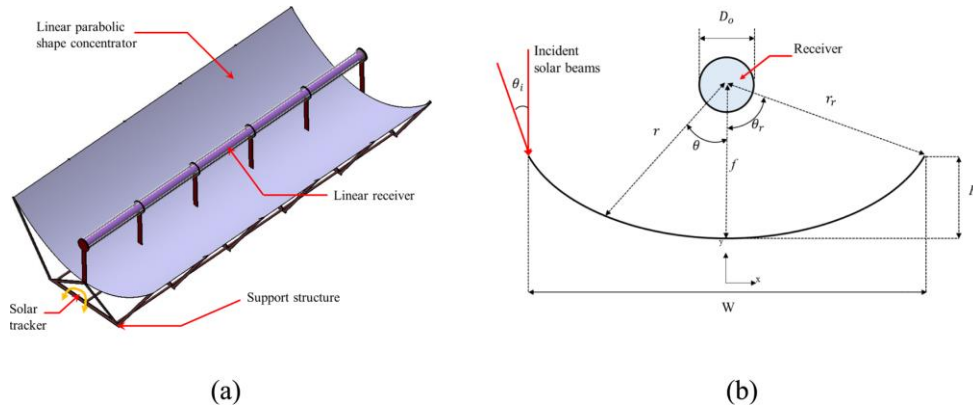


Figure 3: Schematic layout and main components of a parabolic trough solar collector [3].

A PTSC consists mainly of a reflector, supporting structure, receiver, working fluid, and tracking system [1]. Each part affects the optical or thermal behavior of the collector. The reflector must maintain high reflectance and accurate parabolic geometry. The supporting structure must preserve the alignment between the mirror and receiver under self-weight, wind, and operating loads. The receiver must absorb the concentrated radiation while reducing thermal losses. The working fluid must remove heat effectively from the absorber tube. Finally, the tracking system must keep the collector aligned with the Sun so that radiation remains concentrated near the focal line.

Table 2: Main components of a PTSC and their effect on collector performance.

Component	Main function	Relevance to PTSC performance
Parabolic reflector	Reflects and concentrates solar radiation onto the receiver tube	Reflectance, cleanliness, shape accuracy, and slope errors directly affect optical efficiency
Receiver tube	Absorbs concentrated radiation and transfers heat to the HTF	Selective coating, absorber material, diameter, and wall temperature influence heat gain and losses
Glass envelope and annular space	Reduce convection and radiation losses from the absorber to the surroundings	Vacuum quality, glass transmittance, and residual gas affect thermal losses
Supporting structure	Maintains the geometric position of the mirror and receiver	Structural deformation can cause optical misalignment and reduce concentration accuracy
Solar tracking system	Aligns the collector with the Sun during operation	Tracking errors increase incidence losses and reduce useful heat gain
Working fluid	Transports absorbed heat from the receiver to the thermal process	Thermal conductivity, heat capacity, viscosity, and stability affect heat transfer and pumping power

The reflector is one of the most important optical elements of the PTSC. It is commonly made of silvered glass, aluminized reflectors, or polymer-based reflective materials [1]. Tagle-Salazar et al. [1] reported that silvered glass mirrors can reach hemispherical reflectance values up to about 0.96, while aluminized reflectors typically reach values up to about 0.90. High reflectance is desirable because a loss at the mirror directly reduces the energy delivered to the receiver. However, reflectance alone is not sufficient. Dust, abrasion, corrosion, slope errors, tracking errors, and structural deformation can also reduce the amount of radiation intercepted by the receiver [1].

Within the PTSC assembly, the receiver tube is where the actual solar-to-thermal energy conversion takes place. In most commercial designs, this component combines a metallic absorber tube with an outer glass cover. The outer surface of the metal absorber is coated with a selective coating to increase short-wave solar absorptance and reduce long-wave thermal emittance [1,20]. The gap between these two elements is typically evacuated to suppress convective energy loss from the hot absorber surface to the cooler glass wall. Bellows and glass-to-metal seals are used

to accommodate thermal expansion and maintain receiver integrity during operation [1]. This receiver configuration allows the absorber to collect concentrated radiation while limiting heat losses to the environment.

The heat-transfer process in the receiver is not uniform. Part of the absorbed energy is transferred to the HTF through convection at the inner wall of the absorber tube, while the remaining portion can be lost through radiation and convection toward the glass cover and surroundings [18,19]. This makes receiver thermal management a major issue in PTSC design. Forristall [19] showed that PTSC performance is sensitive to optical and thermal losses. His receiver model indicated that thermal performance may decrease by about 15% at an incidence angle of 30° and by about 65% at an incidence angle of 60°. The same work also reported that a reduction of 0.15 in mirror reflectivity can reduce thermal performance by about 24.5%. These numbers show why mirror quality, tracking accuracy, and receiver heat-loss control are not secondary issues; they directly influence the useful thermal output of the collector.

Solar tracking is also central to PTSC operation because parabolic trough collectors depend primarily on direct beam radiation. PTSCs normally use single-axis tracking, either north-south or east-west, depending on the installation strategy and desired seasonal or daily response [1]. The tracking system rotates the collector so that reflected solar radiation remains focused on the receiver tube. A well-designed tracking system improves optical interception, while misalignment increases incidence losses and reduces useful heat gain.

PTSCs are widely used because they offer a practical balance between temperature capability, maturity, and system complexity. They can be applied in industrial process heat, steam generation, solar cooling, desalination, and solar thermal power generation [13,14]. Their operating temperature range is suitable for many medium-temperature industrial processes, and their line-focus design is less complex than point-focus technologies such as solar towers and parabolic dish systems. However, PTSC performance is still limited by receiver heat losses, non-uniform heat-flux distribution, absorber temperature gradients, and the heat-transfer capability of the working fluid.

These limitations explain why recent PTSC research has focused strongly on receiver improvement, advanced working fluids, and internal heat-transfer enhancement. Nazir et al. [18] emphasized that geometrical modifications, corrugated channels, turbulators, and nanofluids are commonly investigated to improve PTSC thermal performance. However, such methods may increase pressure drop, so their value must be judged through hydrothermal performance rather than thermal enhancement alone. This point is important for the present thesis because improving heat transfer inside the PTSC receiver is useful only when the gain in useful heat transfer is not outweighed by excessive pumping-power demand.

1.4. Working Fluids

The working fluid is one of the most important elements in a parabolic trough solar collector because it receives the thermal energy absorbed by the receiver tube and transports it to the storage unit, heat exchanger, steam generator, or direct thermal process. In PTSC literature, the working fluid is commonly referred to as the heat transfer fluid (HTF). Its selection affects the useful heat gain, absorber temperature, pressure drop, pumping power, operating temperature range, and overall thermal efficiency of the system [21]. Therefore, improving the working fluid is not a secondary design issue, but a direct route for improving PTSC performance.

The suitability of a HTF depends on several thermophysical and operational properties. For a working fluid to perform well in a PTSC, it must combine strong heat-carrying capacity with practical handling properties. This means adequate thermal conductivity and specific heat for effective energy absorption, manageable viscosity to keep pumping demands reasonable, resistance to thermal breakdown at operating temperatures, chemical compatibility with receiver materials, and acceptable cost for large-scale deployment. Among these properties, thermal conductivity and specific heat directly govern how well the fluid absorbs and transports energy through the receiver. Viscosity plays an equally important role on the hydraulic side, since fluids with higher resistance to flow demand more pumping energy and produce greater pressure losses along the absorber tube. In addition, the degradation temperature, melting or freezing point, density, thermal expansion, and long-term stability of the fluid influence the design and safe operation of the PTSC field [21].

Conventional working fluids used in PTSC systems include water or steam, synthetic thermal oils, molten salts, gases, and, in some research cases, liquid metals. Water has favorable heat capacity, low cost, and environmental safety. It can also be used in direct steam generation systems, which may reduce the need for intermediate heat exchangers. However, water-based PTSC systems face pressure and phase-change challenges. Krishna et al. [21] reported that approximately 80 bar may be required to maintain water in the liquid phase at high operating temperatures. Scaling inside absorber tubes and thermal storage difficulties also limit its practical use for high-temperature PTSC applications.

Synthetic thermal oils — including products such as Therminol VP-1, Dowtherm A, and Syltherm-type formulations — have been widely adopted in commercial PTSC installations because they remain liquid-phase at the required operating temperatures without demanding the elevated pressures that water-based systems need. They are also compatible with many receiver designs and have been used in commercial solar thermal power plants. However, their temperature stability is limited. Therminol VP-1, for example, is typically used within approximately 12-400 °C. Above its recommended operating range, thermal degradation may occur, producing gases and sludge that can reduce fluid lifetime, increase maintenance requirements, and affect system efficiency [21,22].

When thermal energy storage is a design requirement, molten salt mixtures become an attractive working fluid option. Their relatively low vapor pressure, non-flammable nature, and capability to

operate at temperatures beyond the practical ceiling of synthetic oils make them well suited for high-temperature CSP configurations. Solar salt, commonly based on a $\text{NaNO}_3\text{-KNO}_3$ mixture, has been widely studied for CSP applications. Krishna et al. [21] reported that solar salt typically operates between about 222 °C and 600 °C. Kearney et al. [23] also discussed the engineering potential of molten salts as HTFs in trough solar fields. Their main limitation is the relatively high melting point, which requires freeze protection, preheating, and careful operation during cloudy periods or nighttime shutdowns.

Gaseous working fluids, including air, carbon dioxide, supercritical carbon dioxide, nitrogen, and helium, have also been investigated for high-temperature PTSC applications. Their main advantage is that they avoid freezing problems and can operate at high temperatures without liquid-phase stability limits. However, gases generally have lower density and lower heat capacity than liquid HTFs, which can require larger absorber tubes, higher flow rates, or different receiver configurations to achieve adequate heat transfer. Studies on gas-based PTSC systems have shown their potential, but they remain less common than oil and molten salt systems in conventional trough fields [24,25].

Liquid metals have also been considered because of their high thermal conductivity, low viscosity, and wide operating temperature range. Sodium, for example, can operate over a broad temperature window and offers strong heat-transfer capability [21]. However, liquid metals are difficult to deploy commercially in PTSC systems because of safety concerns, reactivity, corrosion issues, cost, and operational complexity. For this reason, they remain more relevant to advanced research than to conventional PTSC operation.

Table 3: Main categories of working fluids used or investigated for PTSC systems.

Working fluid category	Main advantages	Main limitations	Relevance to PTSC performance
Water/steam	Low cost, high heat capacity, environmentally safe, suitable for direct steam generation	High pressure at elevated temperature, scaling, phase-change control, storage difficulty	Useful for low- to medium-temperature applications and direct steam generation, but challenging for high-temperature receiver operation
Synthetic thermal oils	Mature, compatible with commercial PTSC plants, lower pressure than water at high temperature	Thermal degradation above operating limit, sludge formation, moderate thermal conductivity	Common choice for PTSC systems up to about 400 °C
Molten salts	High-temperature capability, non-flammable, useful for thermal storage, low vapor pressure	High freezing point, preheating demand, corrosion concerns	Attractive for higher-temperature PTSC and storage-integrated systems

Gaseous HTFs	No freezing problem, high-temperature potential, chemically stable options	Low density and heat capacity, larger receiver or higher flow demand	Promising for advanced high-temperature PTSC concepts
Liquid metals	Very high thermal conductivity, low viscosity, broad temperature range	Safety, reactivity, corrosion, cost, limited commercial use	Technically promising but not yet practical for most PTSC plants
Nanofluids and hybrid nanofluids	Improved thermal conductivity, enhanced convective heat transfer, possible optical absorption improvement	Stability, agglomeration, viscosity increase, pressure-drop penalty, cost	Highly relevant to receiver heat-transfer enhancement and the direction of this thesis

In recent years, nanofluids have received strong attention as advanced working fluids for solar thermal collectors. Nanofluids are prepared by suspending solid nanoparticles — typically smaller than 100 nm — within a carrier liquid such as water, oil, ethylene glycol, or molten salt [2,26]. The principal motivation for this approach is that solid materials generally conduct heat far more efficiently than liquids, so incorporating them at the nanoscale can improve the effective thermal properties of the resulting mixture. When the particles are adequately stabilized and distributed throughout the carrier, the mixture can exhibit measurably higher effective thermal conductivity and convective heat-transfer coefficient than the base fluid alone. In PTSC receivers, this can reduce absorber wall temperature, improve heat removal, and increase useful heat gain.

Several studies have reported measurable improvement when nanofluids are used in PTSC receivers. Sokhansefat et al. [27] investigated Al_2O_3 /synthetic oil nanofluid and reported that the heat-transfer coefficient could increase by approximately 14-15% at 5 vol.% nanoparticle concentration. Mwesigye et al. [28] studied synthetic oil- Al_2O_3 nanofluid and showed that nanofluid properties influence the thermal and thermodynamic behavior of the receiver. Other studies have also examined metal oxide nanofluids for medium- and high-temperature PTSC applications, including energy and exergy analysis [29]. These results support the view that modifying the HTF can improve PTSC thermal performance without changing the external collector structure.

However, nanofluids also introduce practical limitations. Increasing nanoparticle concentration can increase viscosity, which raises pressure drop and pumping power. Nanoparticles may also agglomerate, settle, or lose stability at high temperature. Therefore, the best nanofluid is not simply the one with the highest thermal conductivity. It must provide a favorable balance between heat-transfer enhancement, pressure-drop penalty, stability, cost, and compatibility with the receiver materials. Olia et al. [2] emphasized that the pressure-drop increase is one of the main drawbacks of using nanofluids in PTSCs, although it can be managed through optimization of particle concentration and mass flow rate.

Hybrid nanofluids have been proposed as a further development of conventional nanofluids. Instead of using a single nanoparticle type, hybrid nanofluids combine two or more nanoparticle

materials in one base fluid. The aim is to obtain a better balance of thermal conductivity, optical absorption, stability, and rheological behavior than can be achieved by a mono nanofluid. Bellos and Tzivanidis [30] compared mono and hybrid nanofluids in a PTSC and reported an efficiency enhancement of about 1.8% when hybrid nanofluid was used, mainly because of Nusselt number improvement. Minea and El-Maghlany [31] also examined the influence of hybrid nanofluids on PTSC performance and discussed their potential as advanced solar thermal fluids.

For the present thesis, working-fluid improvement is especially relevant because the absorber tube is the location where concentrated solar energy must be transferred efficiently to the flowing fluid. Conventional HTFs provide reliable operation, but their thermal limitations can restrict collector performance. Nanofluids and hybrid nanofluids offer a promising route for increasing heat transfer, but their use must be evaluated carefully because thermal improvement may be accompanied by pressure-drop and stability penalties. This balance between heat-transfer gain and hydraulic cost provides the basis for the next subsection, where passive heat-transfer enhancement techniques are introduced.

1.5. Heat Transfer Passive Techniques

The central aim of receiver-side heat-transfer enhancement in PTSC systems is to raise the rate at which energy deposited on the absorber wall is carried away by the flowing working fluid. Inside the PTSC receiver, the absorber tube experiences concentrated irradiance that is typically stronger on the lower, focal-side surface than on the upper side. When the internal convection is insufficient to dissipate this energy rapidly, the absorber wall temperature rises, radiative and conductive losses escalate, and the temperature difference around the tube circumference introduces mechanical stress. Therefore, improving the internal convective heat transfer inside the receiver tube is an important route for increasing useful heat gain and protecting the receiver structure.

Enhancement techniques for internal convective heat transfer are broadly divided into three categories: active, passive, and compound [32]. Active methods depend on an external energy source — such as vibration, acoustic excitation, or electromagnetic forcing — to disturb the flow. Passive methods achieve the same objective through geometry alone, without drawing on any external power supply. Their mechanisms of action typically include flow disruption through inserts or surface modifications, enhanced lateral mixing across the tube cross-section, thinning of the thermal boundary layer near the heated wall, and in some configurations an increase in the wetted surface area available for heat exchange [32]. Compound methods combine two or more enhancement approaches, such as the use of inserts together with nanofluids or modified receiver geometries.

One practical advantage of passive approaches for PTSC receivers is that they can be installed entirely within the existing absorber tube, leaving the optical and structural components of the solar field unchanged. Depending on their geometry, passive devices promote performance improvement through one or more of the following mechanisms: axial swirl, transverse secondary circulation, localized vortex shedding, flow separation and reattachment at geometric features, or

periodic interruption of the developing thermal boundary layer. Each of these mechanisms raises the convective heat-transfer coefficient at the absorber wall surface. The drawback is that the same disturbances responsible for improved mixing also generate additional flow resistance, which translates to higher friction factor and greater pressure drop along the receiver length. As a result, passive techniques must be judged by their thermal-hydraulic performance rather than by Nusselt number alone [18,32].

A widely adopted metric for assessing passive enhancement is the performance evaluation criterion (PEC), which places the thermal gain and the hydraulic cost on a common basis for direct comparison. It is commonly expressed as [18,32]:

$$PEC = \frac{Nu/Nu_s}{(f/f_s)^{1/3}} \quad (1.2)$$

where Nu and f are the Nusselt number and friction factor of the enhanced receiver, while Nu_s and f_s refer to the corresponding smooth receiver. When the PEC exceeds one, the improvement in heat transfer is large enough to offset the additional pumping energy required, confirming that the modification is thermodynamically worthwhile. This is important because a passive method that increases the Nusselt number but causes excessive pressure drop may not improve the overall system performance.

Among passive techniques, twisted tape inserts are widely studied because they are simple and effective in producing swirl flow. The swirl motion increases fluid mixing and disturbs the thermal boundary layer near the absorber wall. However, their performance depends strongly on twist ratio, tape clearance, perforation, louver angle, and tape-wall spacing. Ghadirijafarbeigloo et al. [33] numerically investigated louvered twisted-tape inserts in a PTSC receiver and reported that perforated louvered twisted tapes can provide strong heat-transfer enhancement compared with a smooth tube. Jaramillo et al. [39] also showed that twisted tapes can improve the thermal behavior of PTSCs, especially when the twist ratio is low and the flow regime is suitable. These studies show that twisted tapes are effective, but their design must be optimized to avoid excessive friction losses.

Perforated plates and porous inserts are another group of passive enhancement devices. They disturb the flow and create jet-like mixing through perforations, which improves heat transfer between the absorber wall and the working fluid. Mwesigye et al. [34] studied centrally placed perforated plate inserts and showed that such inserts can improve both heat transfer and thermodynamic performance. In a later study, Mwesigye et al. [35] examined wall-detached twisted tape inserts and reported significant improvement in receiver performance, including reduced entropy generation. These findings are important because they link passive enhancement not only to first-law thermal efficiency, but also to second-law performance.

Corrugated, ribbed, finned, and wavy absorber tubes enhance heat transfer through direct modification of the absorber geometry. In these designs, the internal wall of the receiver is shaped so that the boundary layer is repeatedly interrupted. The resulting flow separation and reattachment increase local turbulence and improve convective heat transfer. Compared with inserted obstacles, modified absorber walls may offer lower blockage, but they can be more difficult to manufacture and may introduce mechanical or thermal-stress concerns. Liu et al. [38] investigated ribbed absorber tubes and showed that rib arrangement has a strong influence on PTSC receiver performance. Similarly, wavy and corrugated receiver designs have been investigated because they can improve the heat-transfer process while maintaining a moderate pressure-drop penalty [18,32].

Wavy tape inserts and wavy promoters are particularly relevant to the direction of this thesis. Their advantage is that they can generate localized vortices and improve mixing near the heated absorber wall, which is useful under concentrated solar flux. Zhu et al. [37] studied wavy-tape inserts for managing highly concentrated solar energy on the absorber tube of a parabolic trough receiver. Their results showed that wavy inserts can improve heat removal and reduce receiver heat losses. More recently, Mohammed et al. [40] investigated a PTSC receiver equipped with wavy promoters and operated with hybrid nanofluids. They reported that using Fe_2O_3 -GO/Syltherm oil hybrid nanofluid at 2.0% concentration with wavy promoters increased the average Nusselt number by 150.4%, improved thermal efficiency by 18.51%, improved exergetic efficiency by 16.21%, and reduced total entropy generation by up to 48.27%. These results show why passive enhancement and advanced working fluids are often considered together.

The comparative review by Nazir et al. [18] provides an important caution. Their analysis showed that obstacles inserted inside the absorber tube can produce stronger Nusselt-number enhancement than corrugated channels, but they also tend to increase friction factor more severely. For example, their comparison reported that some inserted obstacles can increase average Nusselt number by several times compared with a smooth tube, while some configurations also produce friction-factor ratios greater than ten. In their PEC comparison, inserted rods, centrally placed perforated plate inserts, conical strip inserts, wavy tape inserts, and louvered twisted tapes were among the better-performing configurations, but the hydraulic penalty varied significantly among them [18]. This confirms that passive techniques should not be selected only because they increase heat transfer; their pressure-drop behavior must also be evaluated.

Table 4: Summary of major passive heat-transfer enhancement techniques in PTSC receivers.

Technique category	Examples	Main heat-transfer mechanism	Main limitation	Representative studies
Twisted tape inserts	Twisted tape, louvered twisted tape, wall-detached twisted tape, helical screw-tape	Swirl flow, longer flow path, improved mixing near the absorber wall	Increased pressure drop, strong sensitivity to twist ratio and clearance	[33], [35], [36], [39]

Perforated and porous inserts	Perforated plates, porous discs, porous rings	Flow disturbance, jetting through holes, local vortex formation	Flow blockage and additional friction losses	[34], [18]
Corrugated or ribbed absorber tubes	Ribs, grooves, dimples, helical fins, wavy walls	Boundary-layer interruption, flow separation and reattachment, increased heat-transfer area	More difficult manufacturing and possible thermal-stress concerns	[32], [38]
Wavy tapes and promoters	Wavy tape insert, central wavy promoter	Local swirl, enhanced mixing, improved heat removal in high-flux regions	Additional flow resistance and geometry-dependent performance	[37], [40]
Metal foams and porous media	Full or partial porous inserts, metal foam structures	High surface area, strong fluid-solid interaction, improved conduction paths	Large pressure drop, possible fouling and manufacturing complexity	[18], [32]

Overall, passive techniques provide a practical way to improve PTSC receiver performance because they directly target the internal convective resistance between the absorber tube and the working fluid. Their main benefit is the enhancement of heat transfer without external power input. Their main drawback is the increase in pressure drop. Therefore, the most suitable passive technique is not necessarily the one that produces the highest Nusselt number, but the one that provides the best balance between heat-transfer enhancement and hydraulic penalty. This balance provides the technical link between the present section and the next part of the proposal, where the problem statement is developed around improving PTSC performance through an optimized thermal-hydraulic approach.

1.6. Problem statement

Parabolic trough solar collectors are among the most mature concentrating solar technologies, yet their overall performance remains strongly limited by the thermal behavior of the receiver tube. The receiver is exposed to concentrated solar radiation that is not always distributed uniformly around the absorber circumference. If the absorbed energy cannot be removed quickly enough by the flowing fluid, the absorber surface temperature climbs, radiation losses to the surroundings intensify, and the circumferential temperature distribution around the tube becomes increasingly non-uniform. Such thermal non-uniformity introduces mechanical stress in the absorber material, shortens receiver service life, and raises the risk of permanent deformation or structural failure at the glass-to-metal seal [40,41]. Therefore, the receiver is not only the location where useful energy is collected, but also the region where major thermal losses and structural concerns originate.

Conventional heat transfer fluids provide stable operation in many PTSC systems, but their limitations restrict further performance improvement. Water has high heat capacity and low cost, but high-pressure operation is required at elevated temperatures. Synthetic thermal oils are widely used because of their maturity and compatibility with commercial PTSC systems, yet their thermal stability is usually limited to about 400 °C. Molten salts can operate at higher temperatures and are useful for thermal energy storage, but their high freezing point requires careful thermal management [21-23]. These limitations make the working fluid a central factor in PTSC performance, especially when the objective is to increase useful heat gain while maintaining safe and practical operating conditions.

Advanced working fluids, especially nanofluids and hybrid nanofluids, have been proposed to overcome the limited thermal conductivity of conventional fluids. Several studies have shown that nanofluids can increase the heat transfer coefficient, improve thermal efficiency, and reduce entropy generation in PTSC receivers [2,27-31]. However, this improvement is not obtained without cost. Increasing nanoparticle concentration can raise viscosity, which increases pressure drop and pumping power. For example, Mwesigye et al. [42] reported that using Cu-Therminol VP-1 nanofluid at 6% volume fraction increased the convective heat transfer coefficient by up to 32% and thermal efficiency by 12.5%, but the receiver pressure drop increased by more than 100%. This shows that the use of advanced working fluids must be evaluated as a thermal-hydraulic problem, not only as a heat-transfer improvement.

Passive heat-transfer enhancement techniques face a similar trade-off. Inserts, turbulators, corrugations, ribs, fins, and wavy promoters can disturb the flow, enhance mixing, reduce thermal boundary-layer thickness, and improve heat removal from the absorber wall. These effects can reduce absorber temperature and improve temperature uniformity. However, the same mechanisms that enhance heat transfer also increase flow resistance. For instance, wall-detached twisted tape inserts were reported to improve thermal characteristics by up to 170%, reduce the circumferential temperature difference by about 69%, and decrease entropy generation by about 59%, but such improvements are associated with added hydraulic resistance [35]. Therefore, passive techniques should not be selected only on the basis of Nusselt number enhancement. Their real value depends on whether the heat-transfer gain is large enough to justify the pressure-drop and pumping-power penalties [18,32].

The combined use of advanced working fluids and passive enhancement techniques is promising because it can improve both the thermal properties of the fluid and the flow structure inside the absorber tube. Mohammed et al. [40] demonstrated the potential of this approach by studying a PTSC receiver equipped with wavy promoters and operated with hybrid nanofluids. Their results showed that Fe₂O₃-GO/Syltherm oil hybrid nanofluid at 2.0% concentration increased the average Nusselt number by 150.4%, improved thermal efficiency by 18.51%, improved exergetic efficiency by 16.21%, and reduced total entropy generation by up to 48.27%. At the same time, the friction factor increased by about 22.2-39.1%, and the reported PEC range of 1.24-2.46

confirmed that the performance benefit depends on the balance between heat-transfer enhancement and hydraulic penalty [40].

Table 5: Synthesis of the research problem in PTSC receiver enhancement.

Issue	Technical evidence	Consequence for PTSC performance
Non-uniform heat flux on the receiver	Concentrated solar radiation produces non-uniform absorber heating and circumferential temperature differences [40,41]	Higher absorber temperature, thermal stress, heat loss, and possible receiver reliability problems
Limitations of conventional HTFs	Thermal oils have temperature limits, molten salts require freeze protection, and water may require high-pressure operation [21-23]	Restricted operating range and limited heat-transfer improvement
Nanofluid and hybrid nanofluid trade-off	Nanofluids can improve heat transfer, but increased viscosity can raise pressure drop and pumping power [2,42]	Thermal efficiency may improve, but hydraulic penalty must be controlled
Passive enhancement trade-off	Turbulators and inserts improve mixing and Nusselt number, but also increase friction factor and pressure drop [18,32,35]	Heat-transfer enhancement alone is not enough to prove better system performance
Need for integrated evaluation	Combined enhancement can improve thermal and thermodynamic performance, but depends on Reynolds number, inlet temperature, nanoparticle concentration, and geometry [40]	A validated CFD-based thermal-hydraulic and thermodynamic assessment is required

Based on these limitations, the central problem addressed in this thesis is that the thermal performance of PTSC receiver tubes remains constrained by insufficient heat removal, non-uniform absorber temperature distribution, and the competing effects of heat-transfer enhancement and hydraulic resistance. Conventional working fluids alone are limited by their thermophysical properties, while passive enhancement techniques alone may increase pressure drop. Nanofluids and hybrid nanofluids offer improved heat-transfer capability, but they may also increase viscosity and pumping-power demand. A rigorous numerical study is therefore needed — one that simultaneously assesses the thermal, hydraulic, and thermodynamic consequences of combining an advanced working fluid with a passive insert, and establishes whether the combined approach delivers a measurable performance gain over the unmodified receiver.

Accordingly, this research focuses on assessing PTSC receiver enhancement through a balanced performance framework. The problem is not simply how to increase the Nusselt number, but how to improve useful heat transfer while controlling pressure drop, reducing absorber temperature gradients, improving thermal and exergetic efficiencies, and minimizing entropy generation. This integrated evaluation is necessary to determine whether the proposed enhancement approach provides a real performance advantage over the conventional smooth receiver configuration.

It is important to distinguish the present work from the closely related study of Mohammed et al. [40], which examined a PTSC receiver equipped with wavy promoters using the same Fe_2O_3 -GO/Syltherm 800 hybrid nanofluid, operating range, and performance framework. The present research uses the same validated baseline but introduces a fundamentally different passive geometry: a central hollow twisted square promoter. Unlike a wavy promoter, which applies surface undulations to a solid insert, the hollow twisted square creates two connected fluid passages - an annular region between the absorber wall and the outer promoter surface, and an interior channel through the hollow square core. This dual-passage arrangement alters the velocity distribution, generates a distinct swirl pattern, and modifies the thermal boundary layer in a way that wavy promoters cannot replicate. In addition, the axial twist pitch of the square cross-section is varied systematically in this study, which introduces a new geometric degree of freedom that was not explored in the 2022 work. The present investigation therefore builds directly on the foundation of Mohammed et al. [40] as the next logical step, targeting a promoter geometry that is mechanically distinct, physically different in its flow-enhancement mechanism, and not yet evaluated under the same rigorous thermal-hydraulic and thermodynamic framework.

1.7. Scope and Objectives

The present research focuses on the numerical investigation of heat-transfer enhancement in a parabolic trough solar collector receiver tube. The study is directed toward the combined use of passive enhancement and improved working fluids, since both approaches have been shown to influence PTSC thermal performance, pressure drop, and thermodynamic losses. The main purpose is not only to increase heat transfer, but to determine whether the enhancement produces a real thermal-hydraulic advantage when compared with a conventional smooth receiver.

1.7.1. Scope

The scope of this research is limited to a receiver-scale numerical study of a PTSC absorber tube under turbulent flow conditions. The work will consider a conventional smooth receiver as the baseline case and an enhanced receiver equipped with a selected passive heat-transfer device, such as a central hollow twisted square promoter, as the modified case. The absorber tube will be analyzed under representative solar thermal operating conditions using computational fluid dynamics.

The working-fluid scope includes a conventional Syltherm 800 oil as the base fluid and brick-shaped Fe_2O_3 -GO hybrid nanofluids as improved heat-transfer fluids. The study will evaluate the influence of major operating parameters, including Reynolds number, fluid inlet temperature, and nanoparticle concentration. The planned operating range will cover turbulent Reynolds numbers, with the exact values finalized in the methodology chapter based on the selected geometry, fluid properties, and validation source. A proposed working range for the numerical investigation is $\text{Re} = 5,000$ -100,000, $T_{\text{in}} = 400$ -650 K, and nanoparticle concentration = 0-2 vol.%.

The performance assessment will include thermal, hydraulic, and thermodynamic indicators. The thermal evaluation will consider absorber temperature, outlet temperature, heat-transfer coefficient, and Nusselt number. The hydraulic evaluation will consider friction factor, pressure drop, and pumping-power implications. The thermodynamic evaluation will consider thermal efficiency, exergetic efficiency, entropy generation, and performance evaluation criterion. These indicators are necessary because an enhancement method that increases the Nusselt number may still be unsuitable if it causes excessive pressure drop or entropy generation.

The research does not include experimental fabrication, field testing, long-term nanofluid stability testing, economic optimization, or full solar-field plant simulation. It also does not aim to replace detailed optical design of the entire solar collector. Instead, it focuses on the receiver tube, where the absorbed solar heat is transferred to the working fluid and where the main heat-transfer enhancement mechanisms occur.

1.7.2. Research Aim

The aim of this research is to numerically evaluate whether combining passive heat-transfer enhancement with nano-enhanced working fluids can improve the thermal, hydraulic, and thermodynamic performance of a PTSC receiver tube compared with a conventional smooth receiver operating with a base thermal oil.

1.7.3. Research Objectives

The specific objectives are

- Develop the 3D CFD model to assess how the hollow twisted square promoter (HTSP) affects heat transfer and fluid dynamics in the PTSC absorber tube.
- Investigate the variation in twist pitch Pt (200-400 mm) and hybrid nanofluid parameters (Fe_2O_3 -GO, $\phi = 0-2$ vol.%, $Re = 5,000-100,000$, $T_{in} = 400-650$ K) on thermal and flow behavior.
- Carry out energy, exergy, and entropy analyses to establish the effect of all variables on overall thermohydraulic and thermodynamic performance and identify the optimal pitch configuration and operating conditions.

1.8. Thesis Outline

This thesis proposal is organized into five main chapters, followed by the proposed research timeline and references. The structure is designed to move from the general background of parabolic trough solar collector technology to the specific research problem, then to the numerical methodology required to investigate the proposed enhancement approach.

Chapter 1: Introduction presents the background of the study, introduces solar collectors and parabolic trough solar collectors, discusses working fluids and passive heat-transfer enhancement techniques, and defines the research problem, scope, and objectives.

Chapter 2: Literature Review reviews previous studies related to heat-transfer enhancement in PTSC systems. It covers passive enhancement techniques, advanced heat-transfer fluids, and combined enhancement approaches. The chapter also identifies the main research gaps that motivate the present study.

Chapter 3: Methodology describes the numerical framework adopted in this research. It includes the physical model, governing equations, optical and thermal boundary conditions, CFD code structure, mesh generation and grid-independence study, working-fluid and nanofluid property models, numerical solution procedure, and model validation.

Chapter 4: Results and Discussion will present and analyze the numerical findings of the study. The discussion will focus on thermal performance, hydraulic behavior, thermal efficiency, exergetic performance, entropy generation, and overall performance evaluation.

Chapter 5: Conclusions and recommendations will summarize the main findings of the research, highlight the contribution of the study, and provide recommendations for future work.

2. Literature Review

2.1. Introduction

Parabolic trough solar collectors have been widely investigated because they offer a practical route for converting direct solar radiation into useful thermal energy at medium and high temperatures. Their technical maturity, compatibility with industrial heat and power-generation applications, and established commercial use make them one of the most important concentrating solar technologies. However, their performance remains strongly affected by the thermal behavior of the receiver tube, where concentrated solar radiation is absorbed and transferred to the heat transfer fluid. Any limitation in this heat-transfer process can increase absorber temperature, raise thermal losses, and reduce the useful energy delivered by the collector.

The literature on PTSC performance improvement has developed in several directions. Some studies focus on optical and geometrical design, including reflector materials, receiver arrangement, tracking accuracy, concentration ratio, and absorber coating. Other studies focus more directly on improving the thermal behavior of the receiver by modifying the internal flow structure or improving the thermophysical properties of the working fluid. Recent reviews show that receiver-level enhancement has become a central research direction because the absorber tube is the region where optical energy is converted into thermal energy and where many performance losses originate [1,3].

Heat-transfer enhancement in PTSC receivers can generally be approached through three main routes. The first route is the use of passive techniques, such as fins, ribs, corrugations, twisted tapes, wavy tapes, porous inserts, metal foams, and other turbulators. These methods disturb the flow field, promote mixing, interrupt the thermal boundary layer, and increase the heat-transfer coefficient near the absorber wall. Their main drawback is the possible increase in friction factor, pressure drop, and pumping-power requirement. For this reason, the effectiveness of passive techniques cannot be judged from Nusselt number enhancement alone. It must also be evaluated using hydraulic and thermal-hydraulic indicators such as friction factor, pressure drop, and performance evaluation criterion [18,32].

The second route is the use of advanced heat transfer fluids. Conventional HTFs, such as water, synthetic oils, molten salts, and gases, have different advantages and restrictions related to operating temperature, viscosity, thermal stability, heat capacity, cost, and safety. In recent years, nanofluids and hybrid nanofluids have received attention because they can improve effective thermal conductivity and convective heat transfer. Studies reviewed by Olia et al. [2] show that nanofluids can improve thermal efficiency, heat-transfer coefficient, exergy efficiency, and entropy generation behavior in many PTSC cases. Nevertheless, nanofluids also introduce challenges related to pressure drop, viscosity increase, stability, particle agglomeration, cost, and long-term operation. Therefore, advanced HTFs must be evaluated through both thermal and practical operating criteria [2,21].

The third route is the combined approach, where passive enhancement and improved working fluids are used together. This direction is especially relevant because passive devices modify the flow structure, while advanced fluids modify the heat-transfer capability of the fluid itself. In principle, combining both approaches can produce stronger heat-transfer enhancement than either method alone. In practice, the combined approach may also intensify hydraulic penalties, especially when inserts are used with fluids of higher viscosity. Therefore, the combined approach requires balanced evaluation using Nusselt number, friction factor, pressure drop, thermal efficiency, exergetic efficiency, entropy generation, and PEC.

This chapter reviews previous research according to these three streams. First, the general heat-transfer enhancement framework for PTSC receivers is introduced. Then, passive techniques are reviewed with attention to their mechanisms, performance gains, and hydraulic penalties. After that, advanced HTFs are discussed, including conventional fluids, molten salts, gaseous HTFs, nanofluids, and hybrid nanofluids. The chapter then reviews studies that combine passive enhancement with advanced working fluids. Finally, the research gaps are identified from the reviewed evidence and used to justify the numerical methodology adopted in the present research.

The purpose of this literature review is not only to list previous studies, but to evaluate how each enhancement route contributes to PTSC receiver performance and where limitations remain. Particular attention is given to studies that report measurable thermal, hydraulic, and thermodynamic indicators. This is necessary because the present research is concerned with

balanced performance improvement, not heat-transfer enhancement alone. A technique that increases the Nusselt number but causes excessive pressure drop may not provide a real system advantage. Similarly, a nanofluid that improves thermal conductivity but suffers from high viscosity or stability problems may not be suitable for practical PTSC operation. The literature review therefore builds toward a clear research gap: the need for a validated and systematic numerical assessment of PTSC receiver enhancement that accounts for both thermal gain and hydraulic penalty.

2.2. Heat Transfer Enhancement Techniques

The performance of a parabolic trough solar collector depends strongly on the ability of the receiver tube to transfer absorbed solar energy to the heat transfer fluid. In the receiver, concentrated solar radiation is converted into thermal energy at the absorber surface, then conducted through the absorber wall and transferred by convection to the working fluid. If this heat transfer process is weak, the absorber temperature increases, thermal losses rise, and the useful heat gain decreases. For this reason, improving the heat-transfer process inside the receiver tube has become one of the main research directions in PTSC development.

Heat-transfer enhancement techniques used in PTSC systems may be grouped into three broad categories: active techniques, passive techniques, and compound techniques [32]. Active techniques require external energy input or external forcing, such as vibration, magnetic fields, electric fields, surface oscillation, or mechanical agitation. Although these methods can improve heat transfer in some thermal systems, they are less attractive for PTSC receivers because they increase system complexity and may introduce additional control, power, and maintenance requirements. Passive techniques, in contrast, do not require external power input. They enhance heat transfer through changes in geometry, surface features, inserted elements, or porous structures. Compound techniques combine two or more enhancement methods, such as using a passive insert together with a nanofluid.

In PTSC receiver studies, passive techniques have received considerable attention because they can be applied inside the absorber tube without redesigning the entire solar field. These methods include twisted tapes, wavy tapes, perforated plates, wire coils, rods, ribs, fins, corrugations, porous inserts, metal foams, and wavy promoters. Their purpose is to disturb the flow, promote mixing, generate swirl or secondary flow, interrupt the thermal boundary layer, and increase the heat-transfer coefficient near the absorber wall [18,32]. However, the same flow disturbance that improves heat transfer usually increases friction factor and pressure drop. This means that passive enhancement must be evaluated through a thermal-hydraulic framework, not only through heat-transfer improvement.

A second major route is the improvement of the working fluid. Conventional heat transfer fluids such as water, synthetic oils, molten salts, and gases have been widely used in solar thermal systems. Their performance is controlled by thermophysical properties such as thermal conductivity, specific heat, density, viscosity, thermal stability, freezing point, and pressure

limitations [21]. In many PTSC applications, the conventional fluid limits the rate at which the absorber can remove heat. For this reason, advanced HTFs, including nanofluids and hybrid nanofluids, have been investigated as a way to improve effective thermal conductivity and convective heat transfer. Olia et al. [2] reviewed nanofluid applications in PTSCs and reported that nanofluids can improve thermal efficiency, heat-transfer coefficient, exergy efficiency, and entropy generation behavior in many cases. Their main drawback is the possible increase in pressure drop, especially when nanoparticle concentration and fluid viscosity increase.

The third route is the combined approach, where passive techniques and advanced HTFs are used together. This approach is technically attractive because it targets two different limitations at the same time. Passive devices modify the flow structure inside the receiver, while advanced fluids improve the thermal transport capacity of the working fluid. In principle, this can produce stronger enhancement than either method alone. In practice, the combination may also increase hydraulic resistance, particularly when inserts are used with fluids of higher viscosity. Therefore, combined enhancement must be assessed carefully using heat-transfer, hydraulic, and thermodynamic indicators.

Table 7 summarizes the main heat-transfer enhancement routes considered in this literature review. The table also clarifies the logic of the following subsections. Passive techniques are reviewed first because they directly modify the receiver flow structure. Advanced HTFs are then reviewed because they modify the fluid's ability to absorb and transport heat. Finally, combined approaches are discussed because they represent the closest direction to the present research problem.

Table 6: Framework for reviewing heat-transfer enhancement techniques in PTSC receivers.

Enhancement route	Main mechanism	Main benefit	Main limitation	Role in this review
Active techniques	External forcing such as vibration, magnetic field, electric field, or mechanical agitation	Can increase mixing and heat-transfer rate	Requires external power, control, and added system complexity	Mentioned briefly as part of the general classification
Passive techniques	Flow disturbance, boundary-layer disruption, swirl flow, vortices, increased surface area	Improves Nusselt number and heat removal from the absorber wall	Can increase friction factor, pressure drop, and pumping power	Reviewed in detail in Section 2.2.1
Advanced HTFs	Improved thermal conductivity, heat capacity, optical absorption, or thermal stability	Can improve heat-transfer coefficient and useful heat gain	May suffer from viscosity increase, instability, cost, or high-temperature degradation	Reviewed in detail in Section 2.2.2

Combined approach	Simultaneous use of receiver modification and improved working fluid	Can produce stronger thermal enhancement than a single method	May intensify hydraulic penalty and requires balanced evaluation	Reviewed in detail in Section 2.2.3
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The evaluation of these enhancement techniques must be based on more than one performance indicator. The Nusselt number is useful for measuring convective heat-transfer enhancement, but it does not describe the hydraulic cost. The friction factor and pressure drop are needed to estimate the flow resistance introduced by inserts, corrugations, or higher-viscosity fluids. Thermal efficiency describes the useful energy gain from the receiver, while exergetic efficiency and entropy generation provide a deeper view of thermodynamic irreversibility. For passive and combined methods, the performance evaluation criterion is especially useful because it relates heat-transfer improvement to friction penalty [18,32].

Accordingly, the literature review in the following sections is organized around both mechanism and performance. Studies are not only discussed according to the type of technique used, but also according to the balance between thermal gain and hydraulic penalty. This approach is necessary because a method that improves heat transfer may still be unsuitable if it causes excessive pressure drop, pumping-power demand, or instability. The next subsection therefore reviews passive heat-transfer enhancement techniques in PTSC receivers, with attention to their mechanisms, reported performance gains, and limitations.

2.2.1. *Passive Techniques*

Passive heat-transfer enhancement techniques are widely investigated in parabolic trough solar collector receivers because they can improve convective heat transfer without requiring external power input. In a PTSC receiver, the absorber wall receives concentrated solar radiation and transfers the absorbed heat to the working fluid flowing inside the tube. If the internal convection process is weak, the absorber temperature rises, thermal losses increase, and the temperature distribution around the tube becomes less uniform. Passive techniques address this problem by disturbing the flow field, increasing fluid mixing, interrupting the thermal boundary layer, or increasing the effective heat-transfer area.

In the literature, passive techniques used in PTSC receivers can be grouped into two main categories: surface-integrated modifications and inserted turbulators. Surface-integrated modifications include ribs, fins, dimples, grooves, corrugations, wavy absorber tubes, and modified receiver walls. These techniques are built into the absorber surface itself. Inserted turbulators, in contrast, are additional elements placed inside the absorber tube, such as twisted tapes, perforated plates, rods, wire coils, porous media, metal foams, and wavy tapes. Nazir et al. [18] described these two groups as corrugated channels and channels with obstacles, respectively. This distinction is useful because the two groups enhance heat transfer through similar physical mechanisms but produce different hydraulic penalties.

Surface-integrated modifications enhance heat transfer mainly by forcing repeated flow separation and reattachment near the absorber wall. Corrugations, ribs, dimples, and wavy walls disturb the near-wall region and generate local vortices. This reduces the thickness of the thermal boundary layer and improves energy transport from the absorber wall to the working fluid. However, these modifications may also increase wall shear stress and pressure drop. Their practical value therefore depends on whether the gain in Nusselt number is large enough to compensate for the additional hydraulic resistance.

Early studies on corrugated and finned PTSC receivers showed that moderate improvement could be obtained with limited pressure-drop penalties. Reddy and Satyanarayana [43] numerically examined a porous finned receiver using Therminol VP-1 and reported that a trapezoidal porous insert increased the Nusselt number by up to 13.80% at a mass flow rate of 6.41 kg/s, with a pressure-drop penalty of about 1.7 kPa. Reddy et al. [44] later studied a square porous finned receiver and found that the Nusselt number increased by about 17.5%, while the pressure drop reached about 2.0 kPa. These results indicate that receiver surface modification can improve heat transfer, but the improvement may be modest unless the geometry produces strong mixing.

Helical fins and vortex generators have also been used to improve the internal flow structure. Muñoz and Abánades [45] analyzed internally helically finned tubes for PTSC applications and reported an approximate 18.0% heat-transfer improvement with about 10.0% increase in pressure drop. Cheng et al. [46] studied unilateral longitudinal vortex generators inside a parabolic trough receiver and showed that the mean absorber-wall temperature decreased by about 5.9 K, while the maximum temperature decreased by about 7.1 K. The same study reported a thermal-loss reduction of about 13.6%. These findings are important because they show that passive techniques can improve not only the heat-transfer coefficient, but also wall-temperature uniformity and receiver heat-loss behavior.

More recent studies have explored stronger surface modifications such as ribbed absorber tubes, dimples, protrusions, and wavy geometries. Huang et al. [48] investigated dimples, protrusions, and helical fins in a PTSC receiver, showing the relevance of local surface features for disturbing the near-wall region. Liu et al. [38] examined rib arrangements in a parabolic trough receiver and showed that rib geometry has a strong influence on receiver performance. In the comparative review by Nazir et al. [18], the ribbed absorber tube was identified as one of the strongest corrugated-channel configurations, with an average Nusselt number ratio reported as high as 6.30 relative to the smooth channel. This large improvement confirms the potential of ribbed geometries, but such geometries must still be evaluated through friction factor and PEC because stronger mixing usually comes with higher flow resistance.

Inserted turbulators generally produce stronger flow disturbance than surface-integrated modifications. Twisted tapes are among the most common inserted devices because they generate swirl flow and force the fluid to follow a longer helical path. The swirl motion increases mixing between the core flow and near-wall region. However, the performance of twisted tapes depends

strongly on twist ratio, tape clearance, tape perforation, and tape-wall spacing. A low twist ratio usually produces stronger swirl and higher heat transfer, but it also increases friction losses.

Ghadirijafarbeigloo et al. [33] numerically investigated twisted tape and perforated louvered twisted tape inserts in a PTSC receiver. The perforated louvered twisted tape produced Nusselt numbers that were about 150% higher than the plain tube and about 37% higher than conventional twisted tape. Akbarzadeh and Valipour [32] attributed this improvement to the combined effect of swirl flow from the twisted tape and vortex generation from the louvered fins. The same study reported that the thermal performance of the louvered twisted tape was about 26% higher than that of the conventional twisted tape, with the best performance at $Re = 5000$ and a twist ratio of 2.67. Jaramillo et al. [39] also showed that twisted tape inserts can improve PTSC performance, especially at low Reynolds numbers and when the twist ratio is close to unity.

Wall-detached twisted tapes were investigated by Mwesigye et al. [35] as a way to improve heat transfer while allowing part of the flow to pass between the tape and the absorber wall. Their results showed thermal enhancement factors in the range of 4-68%, while the thermal efficiency increased by about 5-10% for lower flow rates and selected twist ratios. This is a useful design direction because complete blockage of the receiver cross-section may not be necessary to improve heat transfer. Instead, a carefully selected wall clearance can help balance mixing and pressure drop.

Perforated plates, porous inserts, and rods represent another important group of inserted turbulators. These devices generate jets, local recirculation zones, and repeated flow impingement, which increase mixing and heat transfer. Mwesigye et al. [34] investigated centrally placed perforated plate inserts and showed that they can improve heat transfer and thermodynamic performance. In the comparative analysis by Nazir et al. [18], centrally placed perforated plates were reported to increase the average Nusselt number by about 7.12 times compared with an empty tube, while the friction factor ratio reached about 12.75. Despite the large friction penalty, the PEC value was reported as approximately 3.05, indicating that the thermal gain still exceeded the hydraulic penalty under the reviewed conditions.

Wavy tape inserts and wire coils have also shown strong enhancement potential. Zhu et al. [37] designed a wavy-tape insert for managing highly concentrated solar energy on the absorber tube of a parabolic trough receiver. Nazir et al. [18] reported that this configuration increased the average Nusselt number by about 4.17 times and achieved a PEC of about 2.44. Wire coils improve heat transfer by increasing the effective flow path, generating secondary flow, and increasing near-wall turbulence. Diwan and Soni [51] studied wire-coil inserts in PTSC absorber tubes, and the comparative analysis by Nazir et al. [18] reported an average Nusselt number ratio of about 4.02 for this type of insert. However, wire coils can also produce significant friction penalties, so their pitch and wire diameter must be optimized carefully.

Some inserted obstacles produce very high heat-transfer enhancement but also severe friction penalties. Chang et al. [49] investigated inserted rods with molten salt as the working fluid and

reported strong heat-transfer enhancement. In the comparison by Nazir et al. [18], this configuration produced the highest reported average Nusselt number ratio among obstacle-based systems, about 13.00, with a PEC of approximately 6.03. Liu et al. [50] studied conical strip inserts and reported strong thermal-hydraulic behavior; the same comparative review reported a Nusselt number ratio of about 6.50, a friction factor ratio of about 14.00, and a PEC of about 2.70. These results show that obstacles can outperform many corrugated-channel designs, but they must be used carefully because the pressure-drop penalty can become large.

A compact comparison of representative passive techniques is presented in Table 8. The table does not attempt to list every available study. Instead, it highlights studies that are useful for understanding the main mechanisms and trade-offs relevant to the present thesis.

Table 7: Representative passive heat-transfer enhancement studies in PTSC receivers.

Study	Passive technique	Method and fluid	Key reported finding	Main limitation or trade-off
Reddy and Satyanarayana [43]	Porous finned receiver with different corrugation shapes	3D CFD, Therminol VP-1	Trapezoidal porous insert increased Nu by up to 13.80% at 6.41 kg/s	Pressure drop increased by about 1.7 kPa
Reddy et al. [44]	Square porous finned receiver	3D CFD, Therminol VP-1	Nu increased by about 17.5%	Pressure drop reached about 2.0 kPa
Muñoz and Abánades [45]	Internal helically finned tube	CFD, water	Heat transfer improved by about 18.0%	Pressure drop increased by about 10.0%
Cheng et al. [46]	Unilateral longitudinal vortex generators	CFD, Syltherm 800	Mean and maximum wall temperatures decreased by 5.9 K and 7.1 K	Geometry-dependent performance and added flow disturbance
Ghadirijafarbeigloo et al. [33]	Perforated louvered twisted tape	CFD, Behran thermal oil	Nu increased by up to 150% compared with plain tube	Friction factor also increased
Mwesigye et al. [35]	Wall-detached twisted tape	Numerical, thermal oil	Thermal enhancement factor ranged from 4-68%	Hydraulic penalty depends strongly on twist ratio and clearance
Mwesigye et al. [34]	Centrally placed perforated plate	Numerical, Syltherm 800	Average Nu ratio reported around 7.12 , PEC around 3.05	Friction factor ratio reported around 12.75

Zhu et al. [37]	Wavy-tape insert	Numerical, Syltherm 800	Average Nu ratio reported around 4.17 , PEC around 2.44	Added flow resistance and geometry sensitivity
Chang et al. [49]	Inserted rods	Numerical, molten salt	Average Nu ratio reported around 13.00 , PEC around 6.03	High blockage potential and increased pressure drop
Liu et al. [50]	Conical strip inserts	Numerical	Average Nu ratio reported around 6.50 , PEC around 2.70	Friction factor ratio reported around 14.00

The reviewed studies show a consistent pattern. Passive techniques can increase heat transfer substantially, but their value depends on the balance between heat-transfer enhancement and hydraulic penalty. Corrugated and surface-integrated receivers often provide more moderate enhancement with lower friction penalties. Inserted obstacles and turbulators can produce much larger Nusselt number ratios, but they usually cause stronger increases in friction factor and pressure drop. This explains why PEC is more useful than the Nusselt number alone when comparing passive techniques.

From the viewpoint of the present research, wavy tapes, wavy promoters, perforated inserts, and ribbed or corrugated geometries are particularly important because they target the same receiver-level problem: improving heat removal from the absorber wall under concentrated solar flux. However, the literature also shows that the enhancement mechanism cannot be separated from the working fluid. A passive device that performs well with thermal oil may behave differently with molten salt, nanofluid, or hybrid nanofluid because viscosity, density, heat capacity, and thermal conductivity affect both heat transfer and pressure drop. This creates a natural transition to the next section, where advanced heat transfer fluids are reviewed as the second major route for improving PTSC receiver performance.

2.2.2. *Advanced HTFs*

The working fluid in a parabolic trough solar collector controls how effectively the absorbed solar energy is removed from the receiver tube and delivered to the thermal process or power cycle. For this reason, the selection of the heat transfer fluid is not only a material choice, but also a performance decision. The main thermophysical properties that influence PTSC behavior include specific heat capacity, thermal conductivity, density, viscosity, melting point, degradation temperature, thermal expansion, corrosion tendency, and high-temperature stability [21]. These properties affect the heat-transfer coefficient, receiver temperature, pressure drop, pumping power, thermal efficiency, and the feasibility of thermal energy storage.

Conventional HTFs used in PTSC systems include water or steam, synthetic thermal oils, molten salts, and gases. Water has high heat capacity, low cost, and environmental safety, but its use at elevated temperatures requires careful pressure control. Krishna et al. [21] noted that water-operated parabolic trough systems may require pressure levels of approximately 80 bar to maintain

the liquid phase at high operating temperatures. Direct steam generation can simplify the plant layout by removing the intermediate oil loop, but it introduces control difficulties associated with phase change, flow instability, and storage integration.

Synthetic thermal oils have been widely used in commercial PTSC plants because they can operate at moderate to high temperatures with lower pressure than water. Therminol VP-1 and Dowtherm-type fluids are common examples. Their main advantage is operational maturity, but their thermal stability is limited. Krishna et al. [21] reported that Therminol and Dowtherm synthetic oils can withstand temperatures up to about 400 °C, while the plant must remain above about 12 °C to avoid crystallization problems. This operating window is suitable for many PTSC applications, but it restricts the maximum working temperature and therefore limits the achievable power-cycle efficiency.

Molten salts are attractive because they can operate at higher temperatures and can also serve as thermal storage media. Eutectic nitrate salts, including solar salt based on $\text{NaNO}_3\text{-KNO}_3$ mixtures, are widely studied for CSP systems. Krishna et al. [21] reported that solar salt typically operates between about 220 °C and 600 °C. Kearney et al. [23] discussed molten salt as a heat transfer fluid for trough solar fields and highlighted its potential for integrating heat transfer and storage. However, molten salts have high freezing points, which require freeze protection, preheating, and careful operation during startup, shutdown, and cloudy periods. Their corrosion behavior must also be considered when selecting absorber and piping materials.

Gaseous HTFs, including air, nitrogen, carbon dioxide, helium, neon, and argon, have been investigated for high-temperature PTSC operation. Their main advantages are that they are generally stable at high temperatures and do not suffer from freezing or liquid-phase degradation. However, gases have lower density and heat capacity than liquids, so larger flow rates, longer receiver lengths, or larger absorber tubes may be needed to transfer the same thermal load. Bellos et al. [24] compared several gas working fluids and reported that helium provided the best exergetic efficiency among the studied gases, with a value of about 0.4338, mainly because of its high heat-transfer coefficient and low pressure drop. Carbon dioxide was also considered practical because of its availability and acceptable energy and exergy performance [21,24].

Liquid metals have also been proposed as advanced HTFs because they combine high thermal conductivity, low viscosity, and broad operating temperature ranges. Sodium, Na-K eutectic, lead-bismuth eutectic, and gallium-based fluids have received research attention. Krishna et al. [21] reported that liquid metals have promising heat-transfer characteristics, but their use in commercial PTSC plants remains limited because of safety, corrosion, cost, reactivity, and operational complexity. Sodium, for example, has strong heat-transfer potential, but its combustibility in contact with water or moisture has restricted wider application. Therefore, liquid metals remain technically interesting but less mature than oils, molten salts, and gases for PTSC systems.

Nanofluids were introduced as a way to overcome the weak thermal conductivity of conventional fluids. A nanofluid consists of nanoparticles, usually smaller than 100 nm, dispersed in a base fluid

such as water, oil, ethylene glycol, or molten salt [2,26]. The purpose is to improve effective thermal conductivity, heat-transfer coefficient, and in some cases optical absorption. In PTSC receivers, improved thermal transport can reduce absorber wall temperature, increase useful heat gain, and decrease thermal losses. However, the addition of nanoparticles can also increase viscosity, which raises pressure drop and pumping power. This means that the performance of nanofluids must be judged using both thermal and hydraulic indicators.

Several numerical studies have reported measurable improvement when nanofluids are used in PTSC receivers. Mwesigye et al. [42] investigated Cu-Therminol VP-1 nanofluid and reported that at 6 vol.% Cu concentration, the convective heat-transfer coefficient and thermal efficiency increased by up to 32% and 12.5%, respectively. However, the pressure drop at 400 K and $Re \approx 650,000$ increased from nearly 3000 Pa/m to 7734 Pa/m, which is more than a 100% increase. The same study reported a maximum entropy generation reduction of about 30% compared with the base fluid. This result is important because it shows the main strength and weakness of nanofluids at the same time: they improve thermal and thermodynamic performance, but may impose a significant hydraulic penalty.

Other studies show that nanoparticle material and base fluid strongly affect the final performance. Basbous et al. [52] examined metallic and metal-oxide nanofluids in Syltherm 800 at 5 vol.% concentration and reported heat-transfer coefficient enhancements of approximately 36%, 33%, 27%, and 18% for Ag, Cu, CuO, and Al_2O_3 nanofluids, respectively. They also reported a reduction in heat loss of about 21%. Toghyani et al. [53] investigated CuO, SiO_2 , TiO_2 , and Al_2O_3 dispersed in Therminol-55 and reported that Al_2O_3 /Therminol-55 produced the highest overall exergy-efficiency improvement, approximately 11%. These studies suggest that material selection should not be based only on thermal conductivity; density, viscosity, particle stability, and optical behavior must also be considered.

Al_2O_3 -based nanofluids have been studied frequently because of their availability, relatively low cost, and acceptable stability. Nayak et al. [54] reported an approximately 7% enhancement in heat-transfer coefficient using 5 vol.% Al_2O_3 /synthetic oil nanofluid compared with synthetic oil alone. Ferraro et al. [55] also studied Al_2O_3 /synthetic oil and reported that thermal efficacy and pressure drop remained nearly unchanged, while the heat-transfer coefficient increased by up to approximately 60% at 5 vol.% concentration. Khakrah et al. [56] used CFD to study Al_2O_3 /synthetic oil nanofluid in a PTC receiver and reported a 14.3% improvement in thermal efficiency. These results confirm that Al_2O_3 nanofluids can improve receiver heat transfer, although the reported enhancement depends strongly on model assumptions, operating range, and property correlations.

Hybrid nanofluids represent a more recent development. They combine two or more nanoparticle types in one base fluid with the aim of improving thermal conductivity, optical absorption, and stability beyond what is possible with a single nanoparticle. Bellos and Tzivanidis [30] compared mono and hybrid nanofluids in a PTSC and reported that using Al_2O_3 - TiO_2 /oil hybrid nanofluid

produced an efficiency enhancement of about 1.8%, mainly due to improved Nusselt number. Minea and El-Maghlany [31] also investigated hybrid nanofluids for PTSC applications and discussed their potential for improving solar thermal performance. The advantage of hybrid nanofluids is that their properties can be tuned by selecting complementary particles. Their limitation is that stability, preparation, viscosity, and cost become more complex than in mono nanofluids.

Molten salt-based nanofluids are also important for high-temperature PTSC and CSP applications. The main motivation is not only to improve heat transfer, but also to enhance thermal energy storage capacity. Krishna et al. [21] emphasized that doping nanoparticles into molten salts can improve properties such as specific heat, melting point behavior, thermal conductivity, and thermal stability. For example, the review reports that adding nano-silica to nitrate salts can increase decomposition temperature and improve the working-temperature range [21]. This direction is promising because the same fluid can act as both HTF and storage medium. However, molten salt nanofluids still face challenges related to dispersion stability, corrosion, high-temperature cycling, and long-term material compatibility.

A compact comparison of representative advanced HTFs and nanofluid studies is presented in Table 9. The table is not meant to include every available study. It highlights the fluid categories and findings most relevant to the present thesis, especially where heat-transfer improvement is coupled with hydraulic or thermodynamic consequences.

Table 8: Representative studies and performance trends for advanced HTFs in PTSC systems.

Study	Advanced HTF	Method and operating focus	Key reported finding	Main limitation or trade-off
Kearney et al. [23]	Molten salt	Engineering assessment for trough solar field	Demonstrated the technical potential of molten salts as HTFs in trough systems	Requires freeze protection and careful material compatibility control
Bellos et al. [24]	Gaseous HTFs	Energetic and exergetic analysis	Helium achieved the highest exergetic efficiency among studied gases, about 0.4338	Low gas density can require larger absorber dimensions or higher flow demand
Biencinto et al. [25]	Supercritical CO ₂	PTC plant concept with large-aperture collectors	Proposed sCO ₂ as a high-temperature working fluid for advanced solar thermal power cycles	Requires high-pressure design and system-level optimization
Mwesigye et al. [42]	Cu-Therminol VP-1 nanofluid	CFD and thermodynamic analysis	At 6 vol.%, heat-transfer coefficient and thermal efficiency increased by 32% and 12.5%	Pressure drop increased from about 3000 Pa/m to 7734 Pa/m

Basbous et al. [52]	Ag, Cu, CuO, and Al ₂ O ₃ in Syltherm 800	Numerical thermal-performance analysis	Heat-transfer enhancement reached 36% for Ag and 33% for Cu at 5 vol.%	Nanoparticle cost, density, and stability affect practicality
Toghyani et al. [53]	Al ₂ O ₃ /Therminol-55 and other nanofluids	Thermodynamic optimization	Al ₂ O ₃ /Therminol-55 improved overall exergy efficiency by about 11%	Optimization depends on selected power-cycle and property models
Ferraro et al. [55]	Al ₂ O ₃ /synthetic oil	Numerical comparison with conventional working fluids	Heat-transfer coefficient increased by up to about 60% at 5 vol.%	Requires verification of stability and pressure-drop behavior
Bellos and Tzivanidis [30]	Al ₂ O ₃ -TiO ₂ /oil hybrid nanofluid	Thermal analysis of mono and hybrid nanofluids	Hybrid nanofluid improved efficiency by about 1.8%	Hybrid preparation, viscosity, and stability require careful control

The reviewed studies show that advanced HTFs can improve PTSC performance, but they do not remove the need for careful thermal-hydraulic evaluation. Conventional fluids are mature and operationally familiar, but they are limited by temperature range, phase behavior, thermal stability, or storage challenges. Molten salts and gases extend the operating temperature range, yet they introduce freezing, corrosion, pressure, or receiver-size concerns. Nanofluids and hybrid nanofluids can improve heat transfer and reduce thermodynamic losses, but their stability, viscosity, pressure drop, and long-term behavior remain important limitations.

From the perspective of the present research, the most relevant finding is that the working fluid can strongly influence both sides of the performance balance. A fluid with higher thermal conductivity can reduce absorber temperature and increase the Nusselt number, but a fluid with higher viscosity can increase pressure drop and pumping power. This is why advanced HTFs should not be reviewed separately from receiver flow enhancement. The next section therefore examines the combined approach, where advanced working fluids are used together with passive techniques to improve the PTSC receiver while still accounting for the hydraulic penalty.

2.2.3. Combined Approach

The combined approach refers to the simultaneous use of receiver-side passive enhancement and improved working fluids in a parabolic trough solar collector. In this approach, the passive technique modifies the flow structure inside the absorber tube, while the advanced HTF improves the thermal transport capacity of the fluid. This is different from using a passive insert alone or a nanofluid alone. The purpose is to address two major limitations at the same time: weak convective heat transfer near the absorber wall and limited thermophysical properties of conventional working fluids.

The physical logic of the combined approach is straightforward. Passive devices such as fins, corrugations, ribs, twisted tapes, wavy tapes, porous inserts, and promoters disturb the flow and

reduce the thickness of the thermal boundary layer. Nanofluids and hybrid nanofluids, on the other hand, can increase effective thermal conductivity and improve the ability of the fluid to absorb and transport heat. When these two methods are combined, the improved fluid is exposed to stronger mixing and a larger heat-transfer interaction with the absorber wall. In principle, this can increase the Nusselt number, reduce absorber-wall temperature, and improve thermal efficiency more effectively than either method alone [57].

This benefit is not automatic. The same combination that improves heat transfer can also increase pressure drop and pumping-power demand. Passive inserts introduce additional flow resistance, while nanofluids may increase viscosity, especially when nanoparticle concentration increases. Therefore, combined enhancement must be evaluated using thermal, hydraulic, and thermodynamic indicators together. Nusselt number alone is not sufficient. Friction factor, pressure drop, thermal efficiency, exergetic efficiency, entropy generation, and performance evaluation criterion are necessary to decide whether the combined method provides a real improvement.

One of the earlier examples of this direction was presented by Bellos et al. [58], who investigated nanofluids together with a converging-diverging absorber tube. Their study showed that modifying the absorber geometry while using an improved working fluid can enhance PTSC thermal performance. In their reported results, Al_2O_3 -thermal oil nanofluid increased thermal efficiency by about 4.25%, while pressurized water produced an increase of about 6.34%. The mean efficiency enhancement compared with the plain tube was about 4.55%. These results are useful because they show that combining fluid and geometry modification can improve collector performance, but the level of improvement depends strongly on the selected fluid and receiver form.

Longitudinal fins combined with nanofluids have also been investigated as a compound enhancement method. Benabderrahmane et al. [59] studied a PTSC receiver using longitudinal fins and different nanofluids. The reported thermal enhancement ranged approximately from 1.3 to 1.8 times, and the study indicated that Cu and SiC nanoparticles produced stronger heat-transfer improvement than carbon and Al_2O_3 nanoparticles under the tested conditions. This confirms that the combined approach is sensitive to nanoparticle material, not only to the passive geometry.

The influence of absorber geometry and nanofluid selection was further examined by Okonkwo et al. [60]. They investigated the exergetic performance of a PTSC using synthetic oil nanofluids and different absorber geometries. Their results showed that the converging-diverging absorber produced the best exergetic enhancement among the studied cases, with an exergetic improvement of approximately 0.65% using Therminol VP-1 and 0.73% using Al_2O_3 /Therminol VP-1 nanofluid. Although these percentage gains are moderate, they are important because they evaluate improvement from the second-law perspective rather than only from thermal efficiency.

Other studies have compared several absorber geometries under nanofluid operation. Khan et al. [61] assessed different absorber tube geometries for a PTSC using Al_2O_3 /water nanofluid. Their comparison showed that the absorber tube with twisted tape insert achieved the highest thermal efficiency of about 72.26%, followed closely by the tube with internal fins at about 72.10%, while

the smooth absorber tube achieved about 71.09%. This result shows that combined methods may produce measurable improvement, but the margin between different passive geometries can be small. Therefore, geometry selection should consider hydraulic penalty, manufacturability, and operating conditions in addition to thermal efficiency.

Recent work has moved toward more complex combined systems, including grooved absorbers, porous receivers, magnetic nanofluids, hybrid nanofluids, and newly shaped turbulators. Biswakarma et al. [62] studied internally helically V-grooved absorber tubes using nanofluid, showing the continued interest in coupling wall-integrated surface modification with improved fluid properties. Samiezadeh et al. [63] investigated a hybrid oil-Cu-Al₂O₃ nanofluid flowing through a porous receiver tube inside a finned PTSC. Shaker et al. [64] studied Al₂O₃-Syltherm oil nanofluid with a flange-shaped turbulator. Ghanbari et al. [65] examined a PTSC equipped with a combined turbulator and a two-phase magnetic hybrid nanofluid. These studies indicate that the combined approach has expanded from simple nanofluid-plus-insert cases toward more integrated receiver designs.

Mohammed et al. [40] investigated a receiver equipped with wavy promoters and operated with hybrid nanofluids. Their study used Fe₂O₃-GO, Fe₂O₃-SiC, and Fe₂O₃-TiO₂ hybrid nanofluids dispersed in Syltherm 800 under turbulent flow conditions. The Reynolds number range was 5,000-100,000, the inlet temperature range was 400-650 K, and nanoparticle concentration was varied up to 2.0%. Their results showed that Fe₂O₃-GO/Syltherm oil at 2.0% concentration increased the average Nusselt number by 150.4% compared with Syltherm oil. The same study reported a reduction in absorber average outlet temperature of 7-31 °C, a PEC range of 1.24-2.46, thermal efficiency improvement of 18.51%, exergetic efficiency improvement of 16.21%, total entropy generation reduction of 48.27%, and entropy generation ratio reduction of 52.6% [40]. These results show the potential of combining hybrid nanofluids with passive promoters, but they also confirm the need to track pressure-drop and friction effects.

A compact comparison of representative combined-approach studies is presented in Table 10. The table focuses on studies that combine an improved receiver or passive feature with an advanced working fluid. It does not include all available studies, but it captures the main directions relevant to the present research.

Table 9: Representative combined-approach studies in PTSC receiver enhancement.

Study	Combined method	Working fluid	Main reported outcome	Main limitation or research implication
Bellos et al. [58]	Converging-diverging absorber tube with nanofluid	Al ₂ O ₃ -thermal oil and pressurized water	Thermal efficiency improved by about 4.25% with Al ₂ O ₃ -thermal oil and 6.34% with pressurized water	Improvement depends strongly on absorber geometry and working-fluid choice

Benabderrahmane et al. [59]	Longitudinal fins with nanofluids	Al ₂ O ₃ , Cu, SiC and carbon-based nanofluids	Thermal enhancement reported in the range of 1.3-1.8 times	Nanoparticle type strongly affects performance
Okonkwo et al. [60]	Absorber geometry modification with nanofluid	Al ₂ O ₃ /Therminol VP-1	Best exergetic enhancement was about 0.73% for Al ₂ O ₃ /Therminol VP-1 with converging-diverging absorber	Exergy gain was moderate, so hydraulic and thermal losses must be considered
Khan et al. [61]	Different absorber geometries with nanofluid	Al ₂ O ₃ /water	Twisted tape absorber achieved about 72.26% thermal efficiency compared with 71.09% for smooth tube	Performance differences between geometries may be small
Biswakarma et al. [62]	Internally helically V-grooved absorber with nanofluid	Nanofluid	Demonstrated the relevance of combining internal grooves with improved fluids	Requires geometry and fluid-property optimization
Samiezadeh et al. [63]	Porous receiver tube inside finned PTSC with hybrid nanofluid	Hybrid oil-Cu-Al ₂ O ₃ nanofluid	Integrated fins, porous structure, and hybrid nanofluid in a CFD receiver model	More complex geometry increases modeling and practical design requirements
Shaker et al. [64]	Flange-shaped turbulator with nanofluid	Al ₂ O ₃ -Syltherm oil nanofluid	Combined a new turbulator geometry with nano-enhanced thermal oil	Hydraulic penalty should be evaluated carefully
Ghanbari et al. [65]	Combined turbulator with two-phase magnetic hybrid nanofluid	Magnetic hybrid nanofluid	Recent example of coupling turbulator design with magnetic hybrid nanofluid behavior	Two-phase modeling, magnetic effects, and stability increase complexity
Mohammed et al. [40]	Wavy promoters with hybrid nanofluids	Fe ₂ O ₃ -GO, Fe ₂ O ₃ -SiC, and Fe ₂ O ₃ -TiO ₂ in Syltherm 800	Nu increased by 150.4% , thermal efficiency by 18.51% , exergetic efficiency by 16.21% , and total entropy generation reduced by 48.27%	Performance depends on Reynolds number, concentration, particle shape, and friction penalty

The reviewed studies show that the combined approach can produce stronger thermal improvement than single-method enhancement. However, the literature also shows that combined systems are more difficult to judge because several variables change at once. The passive geometry affects turbulence, recirculation, flow separation, and wall shear stress. The working fluid affects thermal

conductivity, viscosity, heat capacity, density, and sometimes optical behavior. When these effects interact, the performance trend may not be linear. A higher nanoparticle concentration may improve heat transfer but also increase viscosity. A stronger turbulator may reduce absorber temperature but also raise pressure drop. A geometry that performs well with thermal oil may not perform the same way with nanofluid or hybrid nanofluid.

For this reason, the combined approach is most meaningful when it is evaluated through a balanced framework. The important question is not only whether the Nusselt number increases, but whether the increase is large enough to justify the hydraulic penalty. The most useful studies are therefore those that report Nusselt number, friction factor, pressure drop, thermal efficiency, exergy efficiency, entropy generation, and PEC together. This is also the point where the literature naturally leads to the research gap. There is still a need for carefully validated numerical studies that examine combined receiver enhancement under realistic turbulent PTSC conditions, while accounting for both heat-transfer improvement and hydraulic cost.

2.3. Research Gap

The reviewed literature shows that PTSC receiver enhancement has been widely studied through passive techniques, advanced heat transfer fluids, and combined approaches. However, several gaps remain directly relevant to the present research.

First, although many passive geometries have been investigated, the available studies do not cover all practical receiver-side configurations. Most attention has been given to common inserts such as twisted tapes, fins, ribs, rods, porous inserts, corrugated walls, and wavy tapes [18,32]. Hollow twisted square promoters, which create a dual-passage flow arrangement and introduce twist pitch as a geometric variable, have not been studied under the same systematic thermal, hydraulic, and thermodynamic framework used in the present work. This is a clear gap, since the hollow geometry differs fundamentally from both solid inserts and wavy promoters in how it distributes the fluid and generates swirl.

Second, the literature on hybrid nanofluids in PTSC receivers is still limited compared with studies on conventional fluids and mono nanofluids. Nanofluids have shown potential to improve heat transfer, but their performance is strongly affected by nanoparticle concentration, viscosity, pressure drop, and stability concerns [2,21]. Hybrid nanofluids require further assessment because their thermal advantage must be balanced against possible hydraulic penalties.

Third, studies that combine passive receiver enhancement with hybrid nanofluids are still not sufficiently systematic. In a combined approach, the passive device changes the flow structure, while the hybrid nanofluid changes the thermophysical behavior of the working fluid. This interaction depends on Reynolds number, inlet temperature, nanoparticle concentration, and geometry. Therefore, more controlled CFD-based studies are needed to identify whether the combined enhancement produces a real net benefit compared with a smooth receiver operating with base thermal oil [40,57].

Finally, most receiver-scale studies do not extend their findings to full plant-scale technical feasibility or economic performance. Such analysis would be useful for judging practical deployment, but it is outside the main scope of the present work. If time permits, this research may briefly discuss the possible plant-scale implications of the numerical findings in terms of pumping-power demand and practical feasibility, without performing a full techno-economic assessment.

Based on these gaps, the present research focuses on a receiver-scale CFD investigation of a smooth PTSC absorber tube and an enhanced tube equipped with a central hollow twisted square promoter. Syltherm 800 is used as the base fluid, while brick-shaped $\text{Fe}_2\text{O}_3\text{-GO/Syltherm 800}$ hybrid nanofluid is considered as the improved working fluid. The study will evaluate Reynolds number, inlet temperature, and nanoparticle concentration effects using thermal, hydraulic, and thermodynamic indicators, including Nusselt number, friction factor, pressure drop, thermal efficiency, exergetic efficiency, entropy generation, and PEC.

Table 10: Concise summary of research gaps addressed in this study.

Gap area	Specific gap	How this research responds
Passive geometry	hollow twisted square promoter configurations with varying twist pitch are less explored than common inserts and corrugations. No study has systematically examined this geometry under the same thermal-hydraulic and thermodynamic framework used here.	Models an enhanced PTSC receiver with a central hollow twisted square promoter and systematically varies the twist pitch from 200 mm to 400 mm to identify the optimal geometric configuration.
Hybrid nanofluid	PTSC studies using hybrid nanofluids are fewer than those using mono nanofluids or conventional HTFs.	Uses brick-shaped $\text{Fe}_2\text{O}_3\text{-GO/Syltherm 800}$ as the enhanced working fluid.
Combined approach	The interaction between passive geometry and hybrid nanofluid is not yet fully systematized.	Evaluates the combined effect under controlled Reynolds number, inlet temperature, and concentration ranges.
Practical implication	Plant-scale feasibility and economic reflection are rarely linked to receiver-scale enhancement.	Discusses pumping-power and feasibility implications only if time permits, without claiming a full economic analysis.

2.4. Summary

This chapter reviewed the main research directions used to improve the performance of parabolic trough solar collector receivers. The literature shows that PTSC enhancement is mainly approached through passive heat-transfer techniques, advanced heat transfer fluids, and combined methods. Each route can improve receiver performance, but each also introduces technical limitations that must be considered carefully.

Passive techniques such as fins, ribs, corrugations, twisted tapes, perforated plates, porous inserts, and wavy tapes improve heat transfer by disturbing the flow, increasing mixing, and reducing the thermal boundary-layer resistance. However, these methods often increase friction factor and

pressure drop. Therefore, their effectiveness must be judged using thermal-hydraulic indicators such as Nusselt number, friction factor, pressure drop, and PEC, rather than heat-transfer enhancement alone [18,32].

Advanced HTFs provide another route for improving receiver performance. Conventional fluids such as water, thermal oils, molten salts, and gases are useful but limited by pressure, stability, freezing point, temperature range, or heat-transfer capability. Nanofluids and hybrid nanofluids can improve thermal conductivity and convective heat transfer, but they may also increase viscosity, pumping power, and stability concerns. This means that fluid selection must balance thermal gain with practical operating limitations [2,21,42].

The combined approach is the most relevant direction for the present research. It uses passive enhancement to improve the flow structure and advanced HTFs to improve heat transport. Reviewed studies show that this approach can produce stronger improvement than either method alone. However, the interaction between geometry, Reynolds number, inlet temperature, nanoparticle concentration, fluid properties, and pressure drop remains complex [40,57]. This confirms the need for a controlled numerical investigation that evaluates thermal, hydraulic, and thermodynamic indicators together.

Overall, the reviewed literature supports the central direction of this thesis. Improving PTSC receiver performance requires more than increasing the Nusselt number. A useful enhancement method should reduce absorber temperature, improve useful heat gain, maintain acceptable pressure drop, improve thermal and exergetic efficiencies, and reduce entropy generation. The next chapter therefore presents the methodology used to model, validate, and assess the proposed PTSC receiver enhancement under defined operating conditions.

3. Methodology

3.1. Introduction

This chapter presents the numerical methodology used to investigate the thermal, hydraulic, and thermodynamic performance of a parabolic trough solar collector receiver tube. The study is based on computational fluid dynamics because the flow and heat-transfer behavior inside a PTSC absorber tube is strongly affected by receiver geometry, non-uniform solar heating, turbulence, working-fluid properties, and internal enhancement structures. CFD provides a practical way to resolve the flow field, wall temperature distribution, pressure drop, and heat-transfer characteristics under controlled operating conditions.

The methodology is developed at the receiver-tube scale. A smooth PTSC receiver is considered as the reference case, while an enhanced receiver equipped with a passive heat-transfer device is used as the modified case. The working fluid is nano-enhanced working fluid, depending on the final scope of the thesis. The numerical model follows the general structure used in previous PTSC CFD studies, especially the methodology of Mohammed et al. [40], where a three-dimensional

PTSC receiver equipped with wavy promoters and hybrid nanofluids was investigated under turbulent flow conditions.

The chapter first describes the physical model and computational domain. The governing equations are then presented, followed by the optical and thermal boundary conditions. The CFD code structure, mesh generation procedure, grid-independence test, working-fluid property models, numerical solution settings, and model validation approach are then explained. Finally, the performance parameters used to evaluate the receiver are defined, including Nusselt number, friction factor, pressure drop, thermal efficiency, exergetic efficiency, entropy generation, and performance evaluation criterion.

3.2. Physical Model

The physical model used in this research consists of a three-dimensional parabolic trough solar collector receiver tube equipped with a central hollow twisted square promoter. The model is limited to the absorber tube and the internal enhancement device. The parabolic reflector, glass envelope, annular vacuum region, and external support structure are not included in the computational domain. Instead, the receiver tube is treated as the region where the absorbed solar energy is transferred to the working fluid. This receiver-scale approach allows the analysis to focus on the internal flow structure, heat-transfer behavior, pressure drop, and thermodynamic performance of the enhanced absorber tube.

Two receiver configurations are considered. The first is a smooth absorber tube, which is used as the baseline case for comparison. The second is an enhanced absorber tube containing a central hollow twisted square promoter. The enhanced configuration is the main design investigated in this study. The comparison between the smooth and enhanced cases is necessary to determine whether the proposed promoter improves heat transfer while maintaining an acceptable hydraulic penalty.

The absorber tube has a total length of 4000 mm, an inner diameter of 66 mm, and an outer diameter of 70 mm. The hollow twisted square promoter is positioned along the centerline of the absorber tube and extends through the full tube length. Its outer square side length is 25 mm, and its wall thickness is 2 mm, giving an internal hollow square opening of approximately 21 mm on each side. The promoter is made of stainless steel because of its high thermal conductivity, low density, and suitability for passive heat-transfer enhancement applications. The absorber tube is assumed to be stainless steel, unless otherwise specified in the final material model.

The promoter is exposed to the working fluid on both its outer and inner surfaces. Therefore, the fluid domain includes two connected flow regions: the main flow passage between the absorber inner wall and the outer surface of the promoter, and the internal hollow passage inside the promoter. This configuration is expected to alter the velocity field, generate secondary flow and swirl-like motion, and increase the contact area between the working fluid and heated solid

surfaces. At the same time, the promoter may increase the pressure drop, so its performance must be evaluated using both thermal and hydraulic indicators.

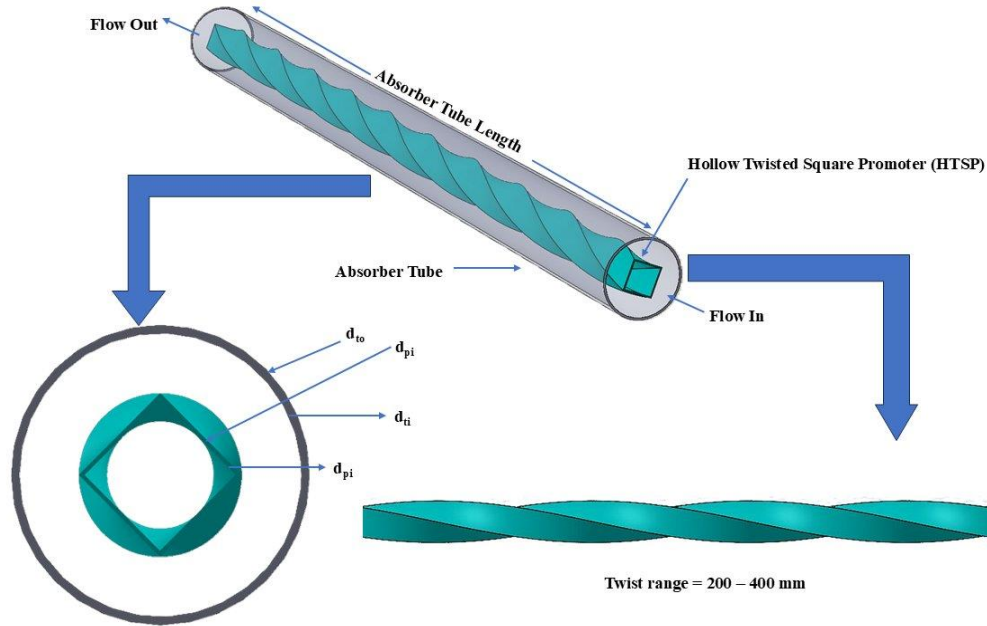


Figure 4: Schematic of the PTSC receiver model showing the absorber tube and hollow twisted square promoter: (a) isometric view, (b) cross-sectional view, and (c) promoter side view with twist pitch range.

The geometry shown in Figure 3.1 is classified in this thesis as a central hollow twisted square promoter rather than a conventional twisted tape. This terminology is used because the device is not a flat tape, but a hollow three-dimensional insert placed at the tube centerline. In the broader passive-enhancement literature, such devices are classified as inserted obstacles or turbulators, which are commonly used to improve mixing and heat transfer inside PTSC absorber tubes [18]. Nazir et al. [18] emphasized that inserted obstacles can improve thermal behavior, but their effectiveness must be assessed together with friction factor and pressure drop.

The twist of the promoter is defined by the axial twist pitch, P_t , which represents the axial distance required for one complete 360° rotation of the square cross-section. In this study, the pitch is varied from 200 mm to 400 mm with an increment of 50 mm. The local twist angle along the tube axis is expressed as:

$$\theta(z) = \frac{2\pi z}{P_t} \quad (3.1)$$

where $\theta(z)$ is the angular rotation of the square cross-section at axial location z , and P_t is the twist pitch. The local coordinates of the twisted square cross-section can be described by the rotation transformation:

$$\begin{bmatrix} x(z) \\ y(z) \end{bmatrix} = \begin{bmatrix} \cos \theta(z) & -\sin \theta(z) \\ \sin \theta(z) & \cos \theta(z) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad (3.2)$$

where x_0 and y_0 are the coordinates of the untwisted square cross-section.

Table 11: Geometrical parameters of the smooth and enhanced absorber tube models.

Parameter	Value
Absorber tube length	4000 mm
Absorber tube inner diameter (d_{ti})	66 mm
Absorber tube outer diameter (d_{to})	70 mm
Absorber tube wall thickness	2 mm
Promoter type	Central hollow twisted square promoter
Promoter outer square side length (d_{po})	25 mm
Promoter wall thickness	2 mm
Promoter internal hollow side length (d_{pi})	21 mm
Promoter length	4000 mm
Twist pitch, P_t	200, 250, 300, 350, and 400 mm
Promoter material	Stainless steel
Baseline case	Smooth absorber tube
Enhanced case	Absorber tube with hollow twisted square promoter

The working fluid cases include pure Syltherm 800 as the base fluid and Fe_2O_3 -GO hybrid nanofluid as the enhanced heat transfer fluid. Brick-shaped nanoparticles with a diameter of 50 nm are considered. The nanoparticle volume concentration is varied from 0 to 2 vol.%. The operating range covers turbulent flow conditions with Reynolds numbers from 5000 to 100000 and inlet temperatures from 400 K to 650 K. These ranges are consistent with the operating conditions commonly used in PTSC receiver studies using Syltherm 800 and hybrid nanofluids.

The numerical investigation is based on the following modeling assumptions:

1. The model is three-dimensional.
2. The flow is steady, turbulent, and incompressible.
3. The working fluid is treated as a single-phase homogeneous fluid.
4. No slip occurs between the nanoparticles and the base fluid.
5. No slip occurs at all solid wall surfaces.

6. The absorber tube and promoter are treated as solid domains.
7. The thermophysical properties of the working fluid may vary with temperature where property correlations are available.
8. Nanoparticle agglomeration, sedimentation, and Brownian slip effects are neglected.
9. The reflector, glass envelope, annular vacuum region, and external air domain are not included.
10. Solar heating is imposed directly on the absorber tube as a thermal boundary condition.
11. The receiver is analyzed at the tube scale rather than the full solar-field scale.

3.3. Governing Equations

The governing equations describe conservation of mass, momentum, and energy for the working fluid flowing inside the absorber tube. Since the receiver operates under turbulent flow conditions, Reynolds-averaged Navier-Stokes equations are adopted. The finite volume method is used to discretize and solve the governing equations in the computational domain.

3.3.1. Continuity equation

For steady incompressible flow, the continuity equation is written as:

$$\frac{\partial(\rho u_i)}{\partial x_i} = 0 \quad (3.3)$$

where ρ is the fluid density and u_i is the time-averaged velocity component in the x_i direction [66,67].

3.3.2. Momentum equation

The Reynolds-averaged momentum equation can be written as:

$$\frac{\partial}{\partial x_j}(\rho u_i u_j) = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \rho u_i' u_j' \right] \quad (3.4)$$

where P is the time-averaged pressure, μ is the dynamic viscosity, and $-\rho u_i' u_j'$ is the Reynolds stress term [66,67].

The Reynolds stresses are modeled using the Boussinesq approximation:

$$-\rho u_i' u_j' = \mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \left(\rho k + \mu_t \frac{\partial u_k}{\partial x_k} \right) \delta_{ij} \quad (3.5)$$

where μ_t is the turbulent viscosity, k is the turbulent kinetic energy, and δ_{ij} is the Kronecker delta [66,67].

3.3.3. Energy equation

The energy equation for the turbulent flow can be expressed as:

$$\frac{\partial}{\partial x_j}(\rho u_j C_p T) = \frac{\partial}{\partial x_j} \left[k_f \frac{\partial T}{\partial x_j} + \frac{\mu_t}{\sigma_{h,t}} \frac{\partial (C_p T)}{\partial x_j} \right] + \Phi \quad (3.6)$$

where C_p is the specific heat capacity, T is temperature, k_f is the thermal conductivity of the fluid, $\sigma_{h,t}$ is the turbulent Prandtl number for energy, and Φ represents viscous dissipation effects [66,67].

3.3.4. Turbulence model

In the reference methodology, the realizable $k - \varepsilon$ turbulence model was used to simulate turbulent transport in the receiver [40]. The transport equation for turbulent kinetic energy is written as:

$$\frac{\partial}{\partial x_j}(\rho k u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k - \rho \varepsilon \quad (3.7)$$

The transport equation for the dissipation rate is:

$$\frac{\partial}{\partial x_j}(\rho \varepsilon u_j) = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \right) \frac{\partial \varepsilon}{\partial x_j} \right] + \rho C_1 S \varepsilon - \rho C_2 \frac{\varepsilon^2}{k + \sqrt{\nu \varepsilon}} \quad (3.8)$$

where G_k is the generation of turbulent kinetic energy, ε is the turbulent dissipation rate, S is the mean strain rate magnitude, and C_1 , C_2 , σ_k , and σ_ε are model constants [66].

The turbulent viscosity is defined as:

$$\mu_t = \rho C_\mu \frac{k^2}{\varepsilon} \quad (3.9)$$

3.3.5. Radiation model

Radiation heat transfer is important in PTSC receivers because the absorber operates at elevated temperature and exchanges radiation with the glass envelope and surroundings. The discrete ordinates radiation model can be written as:

$$\frac{dI(\vec{r}, \vec{s})}{ds} + (a + \sigma_s) I(\vec{r}, \vec{s}) = a n^2 \frac{\sigma T^4}{\pi} + \frac{\sigma_s}{4\pi} \int_{4\pi} I(\vec{r}, \vec{s}') \Phi(\vec{s}, \vec{s}') d\Omega' \quad (3.10)$$

where I is radiation intensity, $\vec{r}$ is the position vector, $\vec{s}$ is the direction vector, a is the absorption coefficient, σ_s is the scattering coefficient, n is the refractive index, σ is the Stefan-Boltzmann constant, and Φ is the phase function [66].

3.3.6. Performance parameters

The Reynolds number is defined as:

$$Re = \frac{\rho u_m d_{ri}}{\mu} \quad (3.11)$$

where u_m is the mean velocity and d_{ri} is the absorber tube inner diameter.

The heat-transfer coefficient is calculated as:

$$h = \frac{q''}{T_{ri} - T_b} \quad (3.12)$$

where q'' is the heat flux, T_{ri} is the average inner wall temperature of the absorber tube, and T_b is the bulk fluid temperature.

The average Nusselt number is defined as:

$$Nu = \frac{h d_{ri}}{k_f} \quad (3.13)$$

The friction factor is calculated using:

$$f = \frac{2\tau_w}{\rho u_m^2} \quad (3.14)$$

where τ_w is the wall shear stress.

The pressure drop along the receiver is expressed as:

$$\Delta P = f \frac{L}{d_{ri}} \frac{\rho u_m^2}{2} \quad (3.15)$$

The useful heat gain is calculated as:

$$\dot{Q}_u = \dot{m} C_p (T_{out} - T_{in}) \quad (3.16)$$

The receiver thermal efficiency is calculated as:

$$\eta_{th} = \frac{\dot{Q}_u - P_p/\eta_{el}}{A_a I_b} \quad (3.17)$$

where $P_p = \dot{V}\Delta P$ is the pumping power, η_{el} is the electrical efficiency factor, A_a is the collector aperture area, and I_b is the incident beam radiation.

The electrical efficiency factor η_{el} accounts for the conversion loss when pumping power is expressed in thermal-equivalent terms. In this study, a value of $\eta_{el} = 0.3$ is adopted, consistent with the approach used by Mohammed et al. [40] and commonly applied in PTSC thermal efficiency evaluations where a typical mechanical-to-electrical conversion efficiency of 30% is assumed [40]. This means that 1 W of pumping power corresponds to approximately 3.33 W of thermal energy input when referred back to the receiver performance balance, ensuring a fair penalty is applied for the hydraulic cost of using passive inserts or high-viscosity nanofluids.

The performance evaluation criterion is used to compare heat-transfer gain with hydraulic penalty. For working-fluid enhancement:

$$PEC_1 = \frac{Nu_{hnf}/Nu_f}{(f_{hnf}/f_f)^{1/3}} \quad (3.18)$$

For passive enhancement:

$$PEC_2 = \frac{Nu_w/Nu_o}{(f_w/f_o)^{1/3}} \quad (3.19)$$

where the subscripts hnf , f , w , and o refer to hybrid nanofluid, base fluid, receiver with passive enhancement, and smooth receiver, respectively.

The total entropy generation rate is expressed as:

$$S_{gen}^T = S_{gen}^F + S_{gen}^H \quad (3.20)$$

where S_{gen}^F is entropy generation due to fluid friction and S_{gen}^H is entropy generation due to heat transfer.

The Bejan number is defined as:

$$Be = \frac{S_{gen}^H}{S_{gen}^T} \quad (3.21)$$

The entropy generation ratio is calculated as:

$$N_s = \frac{(S_{gen}^T)_w}{(S_{gen}^T)_o} \quad (3.22)$$

The exergetic efficiency is evaluated as:

$$\eta_{ex} = \frac{\dot{m}C_p \left[(T_{out} - T_{in}) - T_{amb} \ln \left(\frac{T_{out}}{T_{in}} \right) \right]}{A_a I_b \left[1 + \frac{1}{3} \left(\frac{T_{amb}}{T_s} \right)^4 - \frac{4}{3} \left(\frac{T_{amb}}{T_s} \right) \right]} \quad (3.23)$$

where T_{amb} is the ambient temperature and T_s is the apparent sun temperature.

3.4. Optical Model and Thermal Boundary Conditions

The optical and thermal boundary conditions define how the concentrated solar radiation is applied to the absorber tube and how heat losses occur from the receiver to the surroundings. In a PTSC, the solar flux around the absorber tube is generally non-uniform because the parabolic reflector concentrates radiation mainly on the lower part of the receiver. Therefore, an accurate receiver model should represent the non-uniform circumferential heat-flux distribution as closely as possible.

In the present study, the reflector will not be included as a separate computational domain. Instead, its optical effect will be represented by applying a prescribed solar heat-flux boundary condition to the absorber tube. Following the validated approach used in Mohammed et al. [40], the absorber tube will be subjected to direct and concentrated solar radiation. The upper half of the absorber surface will be treated using a uniform heat-flux boundary condition, while the lower half will be exposed to a non-uniform heat-flux distribution obtained from a Monte Carlo ray-tracing based expression. The direct normal irradiance will be set as:

$$DNI = 1000 \text{ W/m}^2 \quad (3.24)$$

The absorber wall will be modeled as stainless steel with thermal conductivity of approximately $25 \text{ W/m} \cdot \text{K}$.

At the inlet, a velocity boundary condition is applied according to the selected Reynolds number. In the reference paper, Reynolds number ranged from 5000 to 100000, and the inlet fluid temperature ranged from 400 to 650 K [40]. At the outlet, a pressure outlet boundary condition is used. The absorber wall and passive insert surfaces are treated as no-slip walls.

Table 12: Representative boundary conditions and operating parameters.

Parameter	Boundary condition or value
-----------	-----------------------------

Inlet	Velocity inlet based on Reynolds number
Outlet	Pressure outlet
Absorber wall	Solar heat-flux boundary condition
Lower absorber region	Non-uniform heat flux from optical model
Upper absorber region	Uniform heat flux
Absorber and insert walls	No-slip wall
Direct normal irradiance	1000 W/m^2
Ambient temperature	300 K
Inlet temperature range	$400 - 650 \text{ K}$
Reynolds number range	$5000 - 100000$

3.5. CFD Code Structure

The CFD simulation is organized as a sequence of modeling, solution, and post-processing steps. The computational workflow follows the structure used in PTSC receiver studies where geometry creation, mesh generation, material assignment, boundary-condition definition, numerical solution, and validation are performed in sequence.

The geometry is first created for the smooth and enhanced receiver configurations. The enhanced model includes the passive device inside the absorber tube. The computational domain is then divided into fluid and solid regions. The fluid domain represents the working fluid flowing inside the absorber tube. The solid domains represent the absorber wall and passive insert where applicable.

After geometry creation, the mesh is generated with sufficient refinement near walls, inserts, and regions where strong thermal gradients are expected. Inflation layers or near-wall refinement are used to capture wall shear stress and temperature gradients. Materials and thermophysical properties are then assigned to the fluid and solid regions. Temperature-dependent properties may be used for the base fluid and nanofluid when available.

The boundary conditions are then applied, including inlet velocity, inlet temperature, outlet pressure, solar heat flux, no-slip walls, radiation settings, and ambient heat-loss conditions. The solver is then initialized and iterated until convergence is achieved. After convergence, the results are processed to calculate Nusselt number, friction factor, pressure drop, thermal efficiency, entropy generation, exergetic efficiency, and PEC.

The proposed CFD code structure can be summarized as follows:

1. Define receiver geometry and passive enhancement configuration.
2. Generate computational mesh and refine near-wall regions.

3. Assign solid and fluid materials.
4. Define base fluid and nanofluid thermophysical properties.
5. Apply optical and thermal boundary conditions.
6. Select turbulence, radiation, and pressure-velocity coupling models.
7. Solve the governing equations using the finite volume method.
8. Check residuals and monitor thermal and hydraulic variables.
9. Perform grid-independence and model-validation studies.
10. Extract performance parameters for comparison.

3.6. Mesh Generation and Grid Independence Study

The computational mesh was generated using ANSYS Meshing. The fluid domain inside the absorber tube and the hollow interior of the promoter were discretized using an unstructured tetrahedral mesh, with inflation layers applied at all solid boundaries including the absorber inner wall, the outer surfaces of the promoter, and the inner surfaces of the hollow promoter core. The inflation layers were sized to target a dimensionless wall distance of $y^+ \approx 1$, which is required by the enhanced wall treatment used in the turbulence model. To achieve this, the first cell height was calculated from the local wall shear stress estimate based on the inlet velocity and fluid properties at the operating condition, following standard y^+ estimation procedures for turbulent internal flow. This ensured that the viscous sublayer was properly resolved without relying on wall functions, which is important for accurate prediction of the heat-transfer coefficient and friction factor near the absorber wall and promoter surfaces.

Table 13: Representative grid-independence study for PTSC receiver model.

Number of elements	Nu	Pressure drop, ΔP (Pa)	Nu difference (%)	ΔP difference (%)
402,733	178.57	113.53	-	-
1,766,225	196.36	117.21	9.05	3.15
2,464,558	192.19	118.72	2.75	1.27
4,392,771 (adopted)	197.48	118.10	0.56	0.53

The grid-independence study was performed using the enhanced receiver configuration with Syltherm 800 oil at a bulk temperature of 300 K and a uniform heat flux of 1000 W/m². Four mesh densities were tested, ranging from approximately 402,000 to 4,393,000 elements. The Nusselt number and pressure drop were used as the monitoring variables. As shown in Table 3.3, the change in Nusselt number between Mesh 3 (2,464,558 elements) and Mesh 4 (4,392,771 elements) is 0.56%, and the pressure drop difference is 0.00%. These differences fall well below the 1%

threshold adopted as the convergence criterion in this study. Mesh 4 was therefore selected as the adopted mesh for all subsequent simulations, since further refinement does not meaningfully alter the predicted results while adding unnecessary computational cost. The selected mesh consists of 4,392,771 elements and 1,699,142 nodes. The mesh structure, including the inflation layers, cross-sectional arrangement, and axial element distribution, is shown in the figures below.

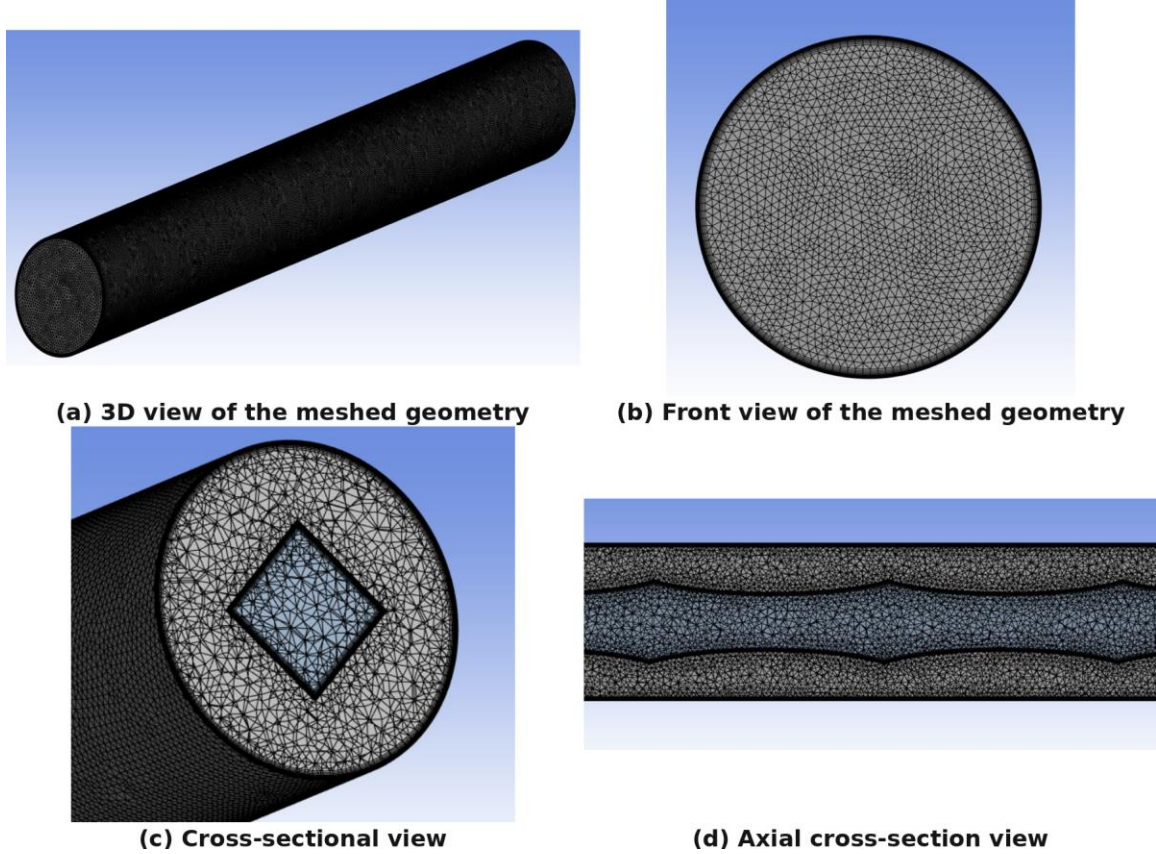


Figure 5: Computational mesh (Mesh 4, 4,392,771 elements): (a) 3D view, (b) front face, (c) cross-section with hollow square promoter, (d) axial section.

3.7. Working and Nano Fluid and Property Models

The working fluid model defines the thermal and hydraulic behavior of the flow inside the receiver. Syltherm 800 oil will be used as the base fluid and hybrid nanofluids of $\text{Fe}_2\text{O}_3\text{-GO}$.

The base-fluid thermophysical properties is temperature-dependent. For Syltherm 800, the thermophysical properties are expressed as polynomial functions of temperature of the form [68]:

$$\psi = a + bT + cT^2 + dT^3 + eT^4 \quad (3.25)$$

where ψ represents the property being calculated, such as dynamic viscosity, thermal conductivity, specific heat capacity, or density. The constants are obtained from the base-fluid property data [68].

For hybrid nanofluids, the effective density is calculated as:

$$\rho_{hnf} = (1 - \phi_p)\rho_{bf} + \phi_{p1}\rho_{p1} + \phi_{p2}\rho_{p2} \quad (3.26)$$

where ρ_{hnf} is the hybrid nanofluid density, ρ_{bf} is the base-fluid density, ρ_{p1} and ρ_{p2} are nanoparticle densities, and ϕ_{p1} and ϕ_{p2} are the volume fractions of the two nanoparticle types.

The effective specific heat capacity is calculated as:

$$(C_p)_{hnf} = \frac{(1 - \phi_p)\rho_{bf}C_{p,bf} + \phi_{p1}\rho_{p1}C_{p1} + \phi_{p2}\rho_{p2}C_{p2}}{\rho_{hnf}} \quad (3.27)$$

The effective thermal conductivity is estimated as:

$$k_{hnf} = k_{bf}(1 + C_k\phi) \quad (3.29)$$

where C_k is a thermal-conductivity shape coefficient. For brick-shaped nanoparticles, this coefficient is taken from the shape-factor data reported by Timofeeva et al. [70], where $Ck = 6.0$ was established for brick morphology based on experimental thermal conductivity enhancement data.

The effective dynamic viscosity is estimated as:

$$\mu_{hnf} = \mu_{bf}[(1 + a\phi_{p1} + b\phi_{p1}^2) + (1 + a\phi_{p2} + b\phi_{p2}^2)] \quad (3.30)$$

where a and b are morphology-dependent coefficients that depend on nanoparticle shape. For brick-shaped nanoparticles, the values are taken from Timofeeva et al. [70], who reported $a = 14.9$ and $b = 123.3$ for this particle morphology based on experimental measurements of viscosity enhancement in alumina nanofluids. These shape-specific coefficients are used consistently throughout the present study to ensure that the viscosity model reflects the actual geometry of the Fe_2O_3 -GO nanoparticles considered.

Table 14: Representative thermophysical properties of Syltherm 800 and selected nanoparticles.

Material	Density (kg/m ³)	Thermal conductivity (W/m.K)	Dynamic viscosity (Pa.s)	Specific heat (J/kg.K)
Syltherm 800	747.2	0.0961	0.00084	1962
Fe ₂ O ₃	5180	6.9	-	670
Graphene oxide	1800	5000	-	717

The nanofluid is assumed to be homogeneous and stable during simulation. Nanoparticles are assumed to be uniformly dispersed in the base fluid, and no slip is assumed between particles and fluid. These assumptions are consistent with a single-phase nanofluid model.

3.8. Numerical Solution

The governing equations are solved numerically using the finite volume method. In the reference methodology, ANSYS Fluent was used as the CFD solver [40]. The finite volume method discretizes the computational domain into control volumes and applies the integral form of the governing equations to each control volume.

A pressure-based steady-state solver is suitable for the incompressible turbulent flow considered in the receiver. Pressure-velocity coupling is handled using the SIMPLE algorithm. The pressure term may be discretized using the PRESTO scheme, especially when pressure gradients and curved flow paths are important. The convective terms in the momentum, energy, turbulence, and radiation equations are discretized using second-order upwind schemes to improve numerical accuracy.

Enhanced wall treatment is used to resolve near-wall gradients and improve wall heat-transfer prediction. This requires sufficiently fine near-wall mesh resolution. The convergence criteria used by Mohammed et al. [40] were 10^{-6} for continuity and 10^{-8} for other variables. In addition to residual convergence, monitored variables such as outlet temperature, average absorber temperature, pressure drop, Nusselt number, and thermal efficiency should also reach stable values.

Table 15: Representative numerical schemes and solver settings.

Item	Setting
Solver	Pressure-based, steady-state
Discretization method	Finite volume method
Pressure-velocity coupling	SIMPLE
Pressure discretization	PRESTO
Momentum discretization	Second-order upwind
Energy discretization	Second-order upwind
Turbulence discretization	Second-order upwind
Radiation model	Discrete ordinates
Wall treatment	Enhanced wall treatment
Continuity residual target	10^{-6}
Other residual target	10^{-8}

The final numerical settings should be selected based on solver stability, residual behavior, and validation quality. If convergence becomes difficult for complex insert geometries or high nanoparticle concentrations, under-relaxation factors may be adjusted while keeping the solution physically consistent.

3.9. Model Validation

Model validation was carried out to establish confidence in the numerical predictions before applying the CFD model to the enhanced receiver configuration. The validation procedure was performed in four stages. First, the smooth absorber tube results were compared with accepted turbulent internal-flow correlations for Nusselt number and friction factor. Second, the model was validated against the experimental LS-2 parabolic trough collector data reported by Dudley et al. [71]. Third, the receiver-scale thermal behavior was compared with the numerical data of Mwesigye et al. [73].

The first validation stage used the smooth receiver operating with Syltherm 800 oil at a bulk temperature of **500 K**. The Reynolds number was varied from **5000 to 100000**, which covers the turbulent range considered in this study. The Nusselt number obtained from the CFD model was compared with the Gnielinski correlation:

$$Nu = \frac{(f/8)(Re - 1000)Pr}{1 + 12.7(f/8)^{1/2}(Pr^{2/3} - 1)} \quad (3.31)$$

where Nu is the Nusselt number, Re is the Reynolds number, Pr is the Prandtl number, and f is the friction factor. The friction factor was validated using the Petukhov correlation:

$$f = (0.79 \ln Re - 1.64)^{-2} \quad (3.32)$$

The comparison showed good agreement between the present CFD results and the correlations. For the Nusselt number, the maximum deviation was approximately **9.04%**, while the average deviation was approximately **7.00%**. For the friction factor, the maximum deviation was approximately **5.18%**, while the average deviation was approximately **1.96%**. These deviations are acceptable for a turbulent receiver-tube CFD model, especially considering that the model is later used for comparative thermal-hydraulic assessment.

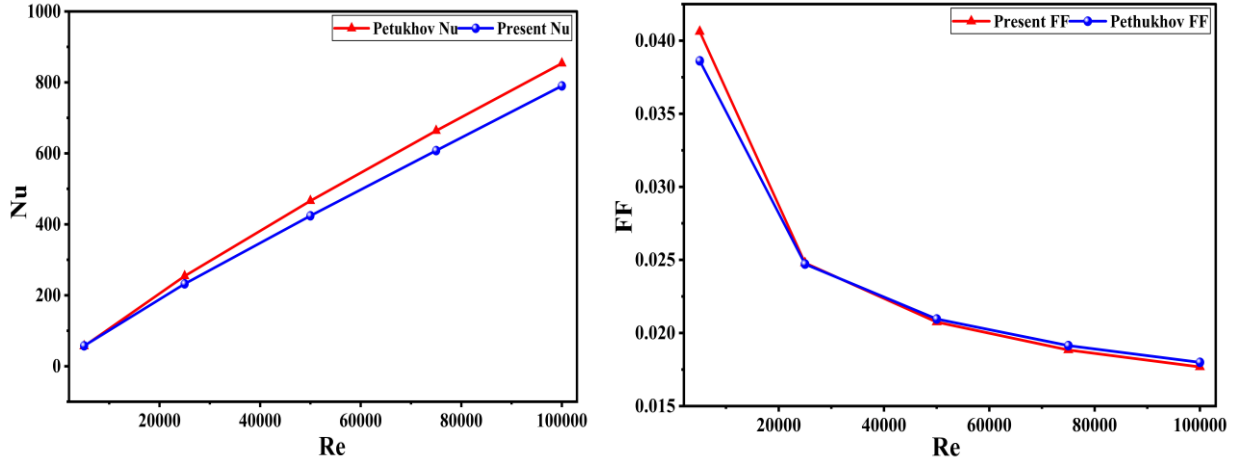


Figure 6: Validation of the average Nusselt number and friction factor against turbulent internal-flow correlations.

The second validation stage was carried out using the experimental LS-2 parabolic trough collector data reported by Dudley et al. [71]. This validation considered the temperature gain across the receiver and the thermal efficiency at different inlet temperatures. The temperature gain was calculated as:

$$\Delta T = T_{out} - T_{in} \quad (3.33)$$

The thermal efficiency for the validation case was calculated as:

$$\eta_{th} = \frac{\dot{m}C_p(T_{out} - T_{in})}{G_b A_{ab}} \times 100 \quad (3.34)$$

where $\dot{m}$ is the mass flow rate, C_p is the specific heat capacity of the working fluid, T_{out} and T_{in} are the outlet and inlet temperatures, G_b is the beam solar radiation, and A_{ab} is the absorber aperture area used in the validation calculation.

The comparison with Dudley et al. [71] showed that the present model reproduced the experimental thermal behavior with reasonable accuracy. The maximum deviation in temperature gain was approximately **9.16%**, while the average deviation was approximately **7.20%**. The thermal efficiency comparison showed stronger agreement, with a maximum deviation of approximately **5.18%** and an average deviation of approximately **1.82%**. This confirms that the model can predict the overall thermal response of the receiver with acceptable accuracy.

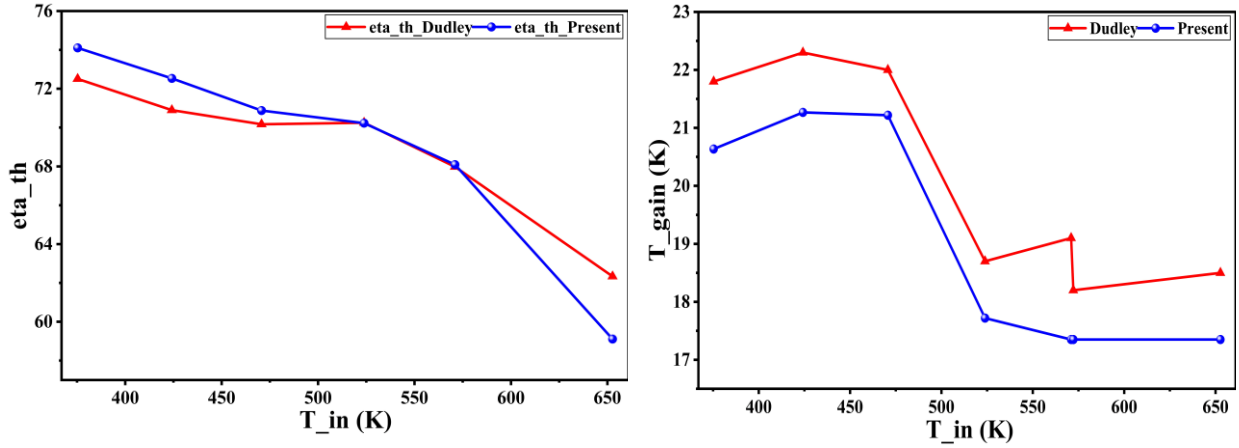


Figure 7: Validation of temperature gain and thermal efficiency against Dudley et al. [71].

The third validation stage compared the present receiver-scale results with the numerical data of Mwesigye et al. [73]. This comparison was used mainly to verify the trend and order of magnitude of the receiver heat-loss behavior. The heat-loss comparison followed the expected increasing trend with absorber-to-ambient temperature difference. This stage was treated as receiver-scale thermal validation because the modeled receiver details and operating assumptions are not identical.

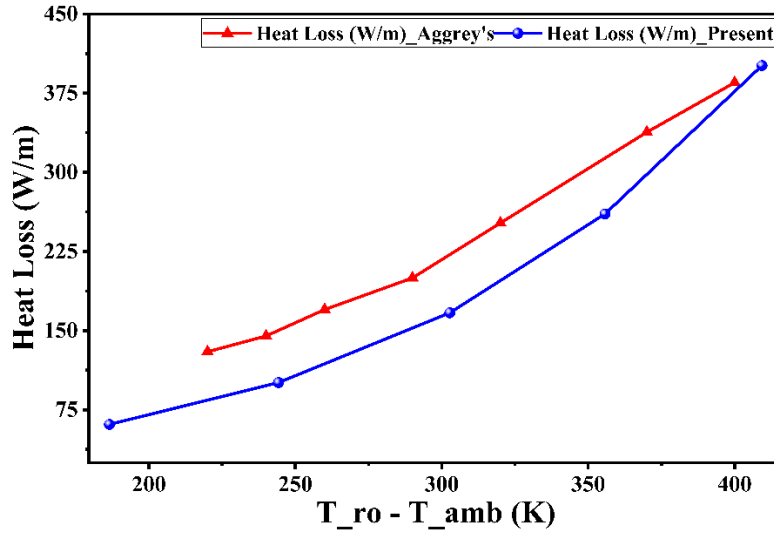


Figure 8: Receiver-scale validation of heat-loss trend against Mwesigye et al. [73].

The fourth validation stage was performed against the numerical results of Sheikholeslami et al. [74], who investigated a parabolic trough collector with a wavy absorber tube and CuO/oil nanofluid using a two-phase mixture model in ANSYS Fluent. Their study examined eight cases combining two Reynolds numbers ($Re = 5,000$ and $20,000$), two nanoparticle volume concentrations ($\phi = 0.01$ and 0.05), and two pitch numbers ($P = 0$ and $P = 0.4$ m). The turbulence was modelled using the RNG $k-\epsilon$ approach with enhanced wall treatment, and the heat flux

distribution on the absorber outer wall was obtained using SolTrace based on a Monte Carlo ray-tracing method. This study is relevant to the present work because it uses the same CFD platform, the same turbulence modelling approach, and examines a geometrically modified absorber tube - a wavy sinusoidal absorber - under turbulent nanofluid flow conditions that are comparable to those of the present model. The friction factor and heat transfer coefficient reported in Sheikholeslami et al. [74] for cases 1-4 ($P = 0$, smooth-equivalent receiver) and cases 5-8 ($P = 0.4$, enhanced receiver) serve as the reference dataset for this stage. The present model's predictions of friction factor and heat transfer coefficient were compared against the reported values at matching Reynolds numbers and concentration conditions. As shown in Figures 3.8(a) and 3.8(b), the present results follow the same trends reported by Sheikholeslami et al. [74]: friction factor decreases with increasing Reynolds number, and heat transfer coefficient increases substantially as Reynolds number rises from 5,000 to 20,000. The maximum deviation in friction factor was approximately **23.6%** and in heat transfer coefficient approximately **30.4%**, which are larger than the deviations observed in Stages 1-3.

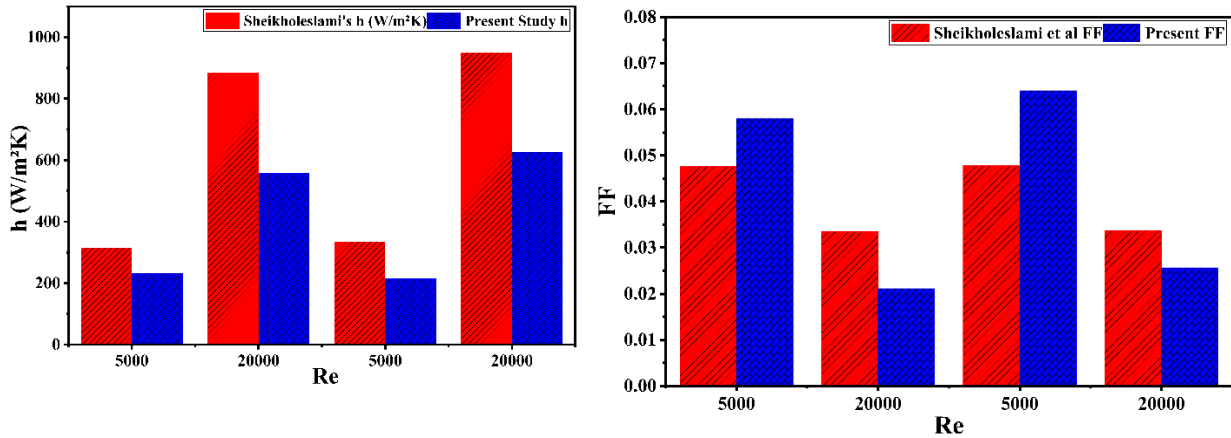


Figure 9: Hydrothermal trend validation of the enhanced receiver against the smooth-receiver results of Sheikholeslami et al [74].

These differences are expected and physically justified because the geometry, fluid, operating temperature, and tube length in Sheikholeslami et al. [74] differ from the present model - their tube is 2 m long with a wavy sinusoidal wall and CuO/oil at 25 °C inlet, while the present model is 4 m long with a smooth tube and Syltherm 800 at 400-650 K. This stage is therefore treated as a hydrothermal trend validation rather than a direct quantitative comparison. Its purpose is to confirm that the present model reproduces the physically correct response to changes in Reynolds number and geometry modification - stronger heat transfer at higher Re , and higher friction factor with geometrical complexity - consistent with the behavior documented in the reference study. It should be noted that the geometry, promoter type, and exact mesh configuration between the present model and Mohammed et al. [40] are not identical, since that study used wavy promoters while this work uses a hollow twisted square promoter. For this reason, the comparison is carried

out at the smooth-receiver level, where both models share the same geometry and boundary conditions, making the deviation a direct reflection of solver and discretization consistency rather than geometry-induced differences. The results confirm that the present numerical setup is consistent with the validated framework of the reference study and can therefore be trusted for comparative assessment of the enhanced receiver.

Table 16: Summary of validation cases for the numerical model.

Validation stage	Reference source	Compared parameter	Range or condition	Main purpose
Smooth-tube correlation validation	Gnielinski and Petukhov correlations	Nusselt number and friction factor	$Re = 5000 - 100000$, Syltherm 800, $T_b = 500K$	Validate the basic turbulent internal-flow behavior
Experimental validation	Dudley et al. [71]	Temperature gain and thermal efficiency	LS-2 parabolic trough collector test cases	Verify receiver-scale thermal response against experimental data
Numerical receiver validation	Mwesigye et al. [73]	heat-loss trend	Receiver-scale thermal comparison	Confirm thermal trend and order of magnitude against published numerical data
Enhanced-geometry trend validation	Sheikholeslami et al. [74]	Heat-transfer coefficient and friction-factor behavior	Smooth-receiver Nu and f comparison at matching Re and T_{in} with Fe_2O_3 -GO/Syltherm 800	Confirm expected hydrothermal response of wavy/enhanced receiver geometry

Overall, the validation results indicate that the CFD model provides acceptable agreement with empirical correlations and published PTSC benchmark data. The model also reproduces the expected hydrothermal behavior of enhanced receiver geometries reported in the literature. Therefore, the validated model is considered suitable for the main objective of this thesis: comparing the smooth receiver with the enhanced receiver equipped with a central hollow twisted square promoter under the same numerical framework.

3.10. Summary

This chapter presented the numerical methodology for evaluating the performance of a PTSC receiver tube. The physical model includes the absorber tube, glass envelope, working-fluid domain, and passive enhancement device. The governing equations include conservation of mass, momentum, and energy, together with turbulence and radiation models. The optical and thermal boundary conditions define the solar heat input, receiver heat losses, inlet flow conditions, and wall constraints.

The methodology also described the CFD workflow, mesh generation procedure, grid-independence study, working-fluid and nanofluid property models, numerical schemes, and validation strategy. The performance parameters include Nusselt number, friction factor, pressure drop, thermal efficiency, exergetic efficiency, entropy generation, and PEC. These indicators allow the receiver to be evaluated not only by heat-transfer enhancement, but also by hydraulic and thermodynamic performance.

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